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MIL-M-28787/367(SH)
13 December 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1553 BUS CONTROLLER MODULE, KEY CODE UKS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1553 bus controller module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.8700 failures per million operating hours for class I or 8.5898 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be UKS.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 115 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	II	S	10	INF	10	INF	10	INF	MOhm
A2-A7	Isolation, pin to pin	II	X	10	INF	10	INF	10	INF	MOhm
B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests----------						
D1-D17	Ibd	V	S	0	100	0	100	0	100	uA
D18,D19	Ibd	V	S	0	1000	0	1000	0	1000	uA
D20-D39	Ibd	V	S	0	100	0	100	0	100	uA
E1-E39	Vio	VI	S	-1.5	-0.3	-1.5	-0.3	-1.5	-0.3	V
F1-F4	Ios	VII	S	-130	-30	-130	-30	-130	-30	uA
F5-F24	Ios	VII	S	-225	-40	-225	-40	-225	-40	uA
F25-F30	Ios	VII	S	-130	-30	-130	-30	-130	-30	uA
F31-F43	Ios	VII	S	-100	-20	-100	-20	-100	-20	uA
F44	Ios	VII	S	-225	-40	-225	-40	-225	-40	uA
G1-G17	Iih	VIII	X	-2	20	-2	20	-2	20	uA
G18	Iih	VIII	X	-2	25	-2	25	-2	25	uA
G19	Iih	VIII	X	-2	50	-2	50	-2	50	uA
G20-G22	Iih	VIII	X	-2	20	-2	20	-2	20	uA
G23,G24	Iih	VIII	X	-200	-50	-200	-50	-200	-50	uA
G25	Iih	VIII	X	-2	20	-2	20	-2	20	uA
G26	Iih	VIII	X	-200	-50	-200	-50	-200	-50	uA
G27-G37	Iih	VIII	X	-2	20	-2	20	-2	20	uA
G38	Iih	VIII	X	-2	100	-2	100	-2	100	uA
G39	Iih	VIII	X	-2	150	-2	150	-2	150	uA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H8	Iil	IX	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
H9-H22	Iil	IX	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
H23,H24	Iil	IX	X	-0.8	-0.2	-0.8	-0.2	-0.8	-0.2	mA
H25	Iil	IX	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
H26	Iil	IX	X	-0.8	-0.2	-0.8	-0.2	-0.8	-0.2	mA
H27-H37	Iil	IX	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
H38,H39	Iil	IX	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
J1-J43	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J44	Voh	X	X	4.1	4.4	4.1	4.4	4.1	4.4	V
J45	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K24	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K25-K38	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K31-K43	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K44	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K45	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L7	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L8-L11	Iozh	XII	X	-225	-100	-225	-100	-225	-100	uA
L12	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L13-L16	Iozh	XII	X	-225	-100	-225	-100	-225	-100	uA
L17-L24	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L25	Iozh	XII	X	-750	145	-750	145	-750	145	uA
L26	Iozh	XII	X	-100	100	-100	100	-100	100	uA
L27	Iozh	XII	X	-350	125	-350	125	-350	125	uA
L28	Iozh	XII	X	-1950	205	-1950	205	-1950	205	uA
L29	Iozh	XII	X	-1950	225	-1950	225	-1950	225	uA
L30	Iozh	XII	X	-2950	285	-2950	285	-2950	285	uA
M1-M7	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M8-M11	Iozl	XIII	X	-350	-200	-350	-200	-350	-200	uA
M12	Iozl	XIII	X	-20	20	-20	20	-20	20	uA

- See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
M13-M16	Iozl	XIII	X	-350	-200	-350	-200	-350	-200	uA
M17-M24	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M25	Iozl	XIII	X	-750	145	-750	145	-750	145	uA
M26	Iozl	XIII	X	-100	100	-100	100	-100	100	uA
M27	Iozl	XIII	X	-350	125	-350	125	-350	125	uA
M28	Iozl	XIII	X	-1950	205	-1950	205	-1850	205	uA
M29	Iozl	XIII	X	-1950	225	-1950	225	-1950	225	uA
M30	Iozl	XIII	X	-2950	285	-2950	285	-2950	285	uA
N1	Icc	XIV	X	250	1000	250	1000	250	1000	mA
N2	Icc	XIV	X	2	25	2	25	2	25	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
R1	PSFC	XVII	S	4.9	9.1	4.9	9.1	4.9	9.1	uF
R2	PSFC	XVII	S	0.007	0.013	0.007	0.013	0.007	0.013	uF
S1,S2	PSTR	XVIII	Q	0	400	0	400	0	400	mV
T1-T43	Output Crosstalk, high level	XIX	Q	0	400	0	400	0	400	mV
U1-U43	Output Crosstalk, low level	XX	Q	0	400	0	400	0	400	mV
V1-V38	Cin	XXI	Q	2	40	2	40	2	40	pF
V39	Cin	XXI	Q	2	75	2	75	2	75	pF
W1-W5	Oscillator	XXII	S	See table XXII						

- See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test tolerance table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/367-1 for class I and M28787/367-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.6.8.1 Electrostatic Discharge (ESD) sensitive marking. The module shall be marked with an electrostatic discharge sensitive symbol as specified in MIL-STD-1389.

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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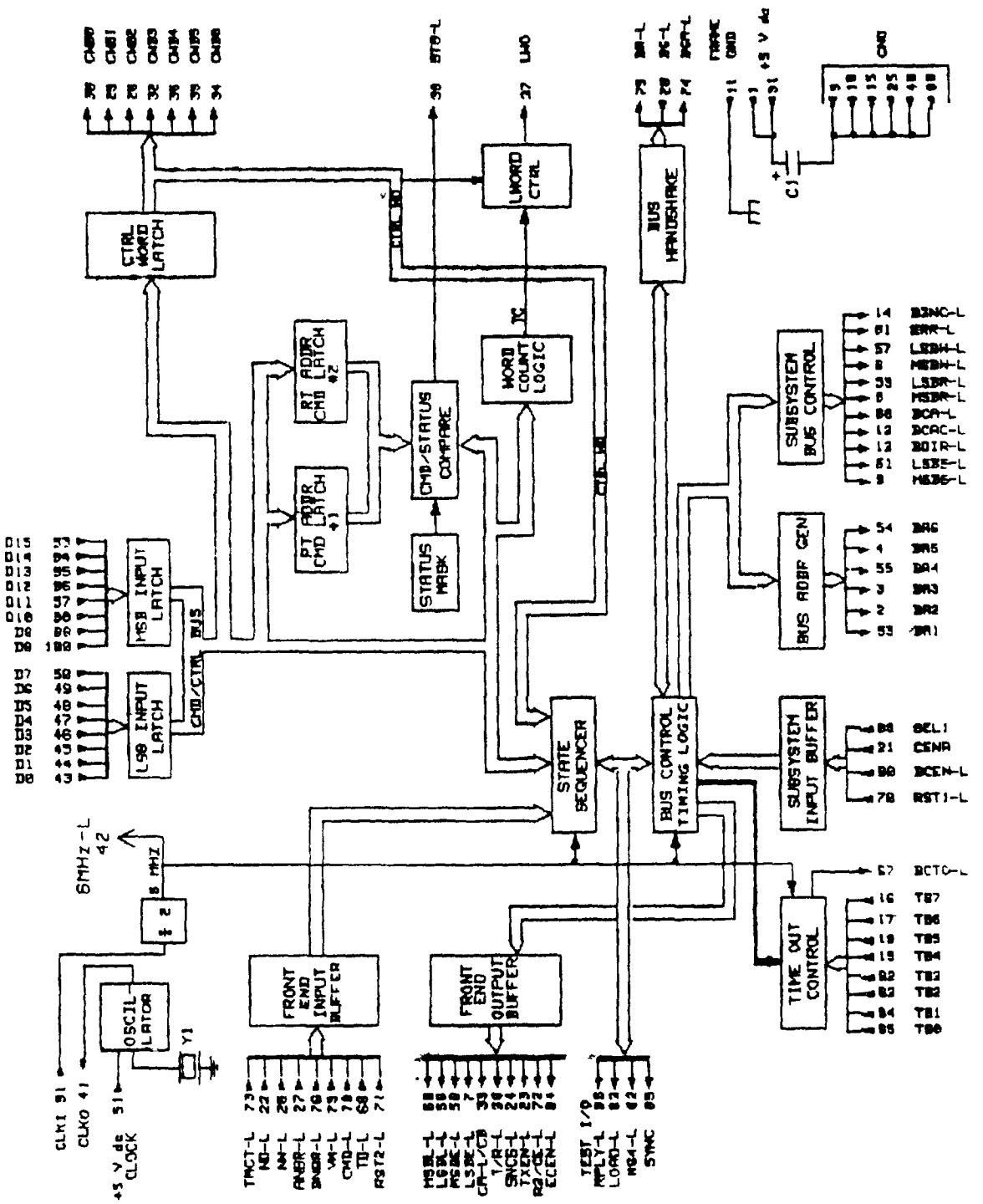


FIGURE 1. UKS functional block diagram.

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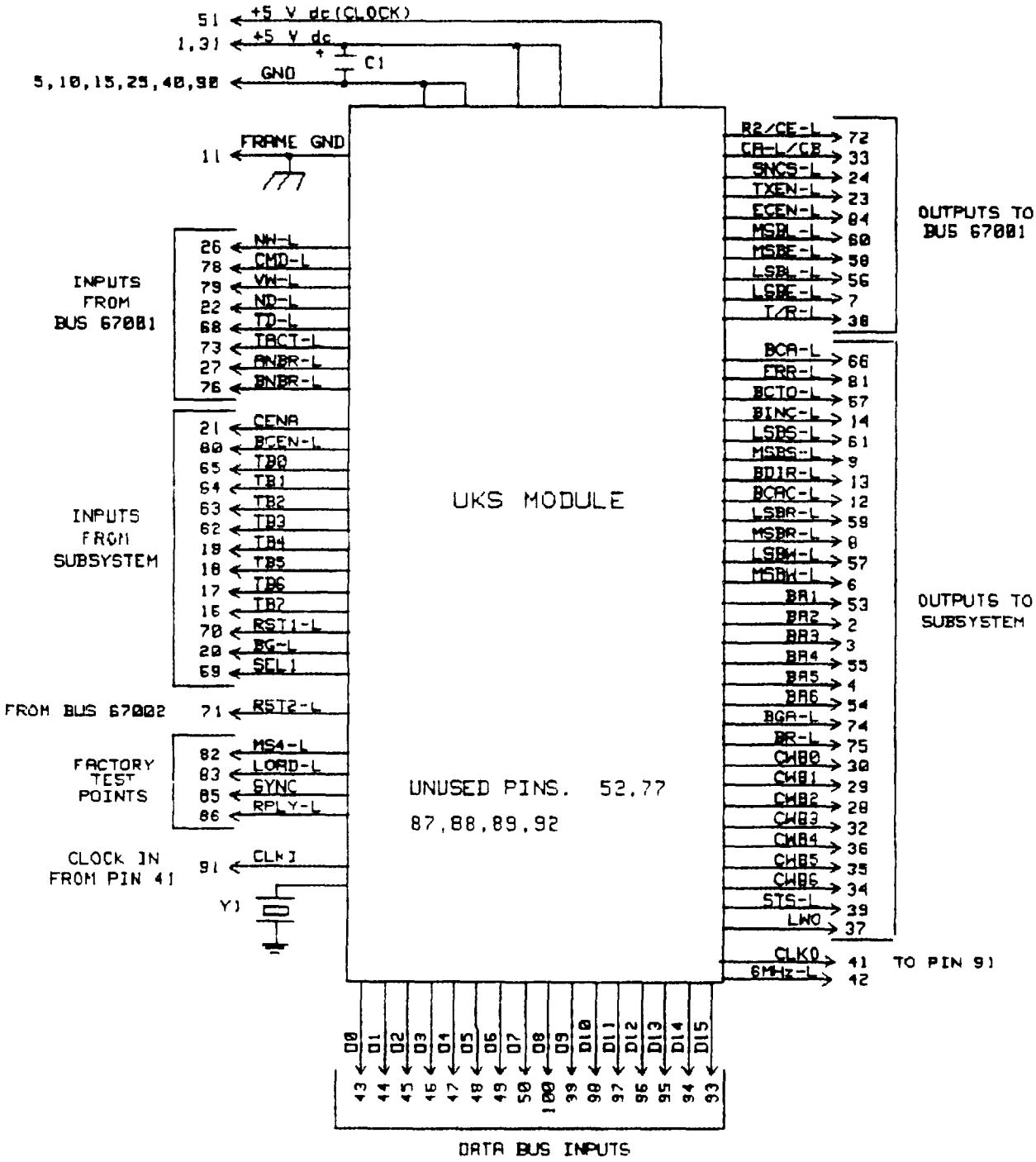


FIGURE 2. UKS pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and G through U shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

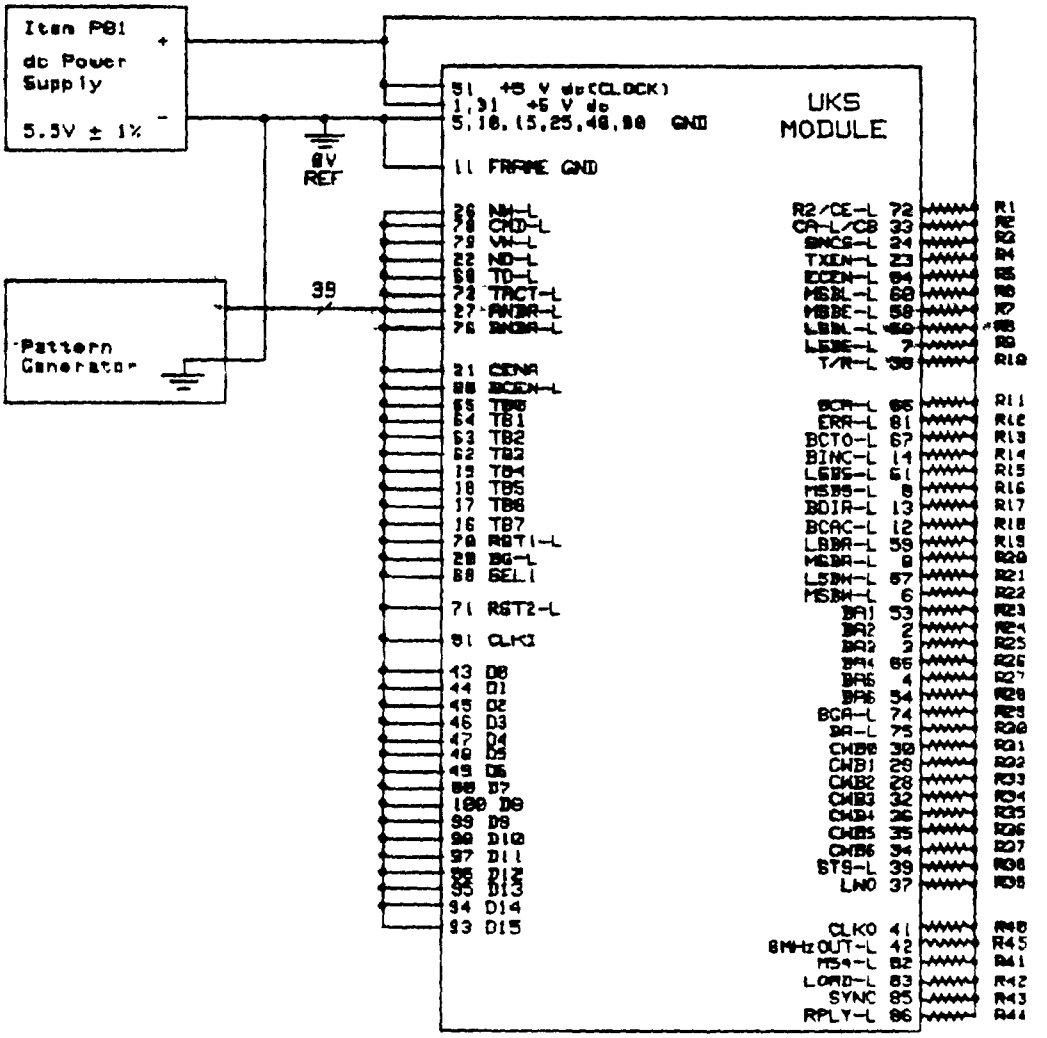
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B and G through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B and G through U shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTES:

1. Set the pattern generator to a pattern rate of 1000 nS ± 200 nS, 50 percent duty cycle using appropriate patterns to exercise the module.

2. R1, R11, R13, R14, R31-R39, R45 = 1.3 KOhm ± 5 percent, 1/4 W. R2-R4, R6-R10, R15-R30 = 432 Ohm ± 5 percent, 1/4 W. R5, R12, R41-R44 = 634 Ohm ± 5 percent, 1/4 W. R40 = 511 Ohm ± 5 percent, 1/4 W.

FIGURE 3. UKS operating temperature, component temperature and life test circuit.

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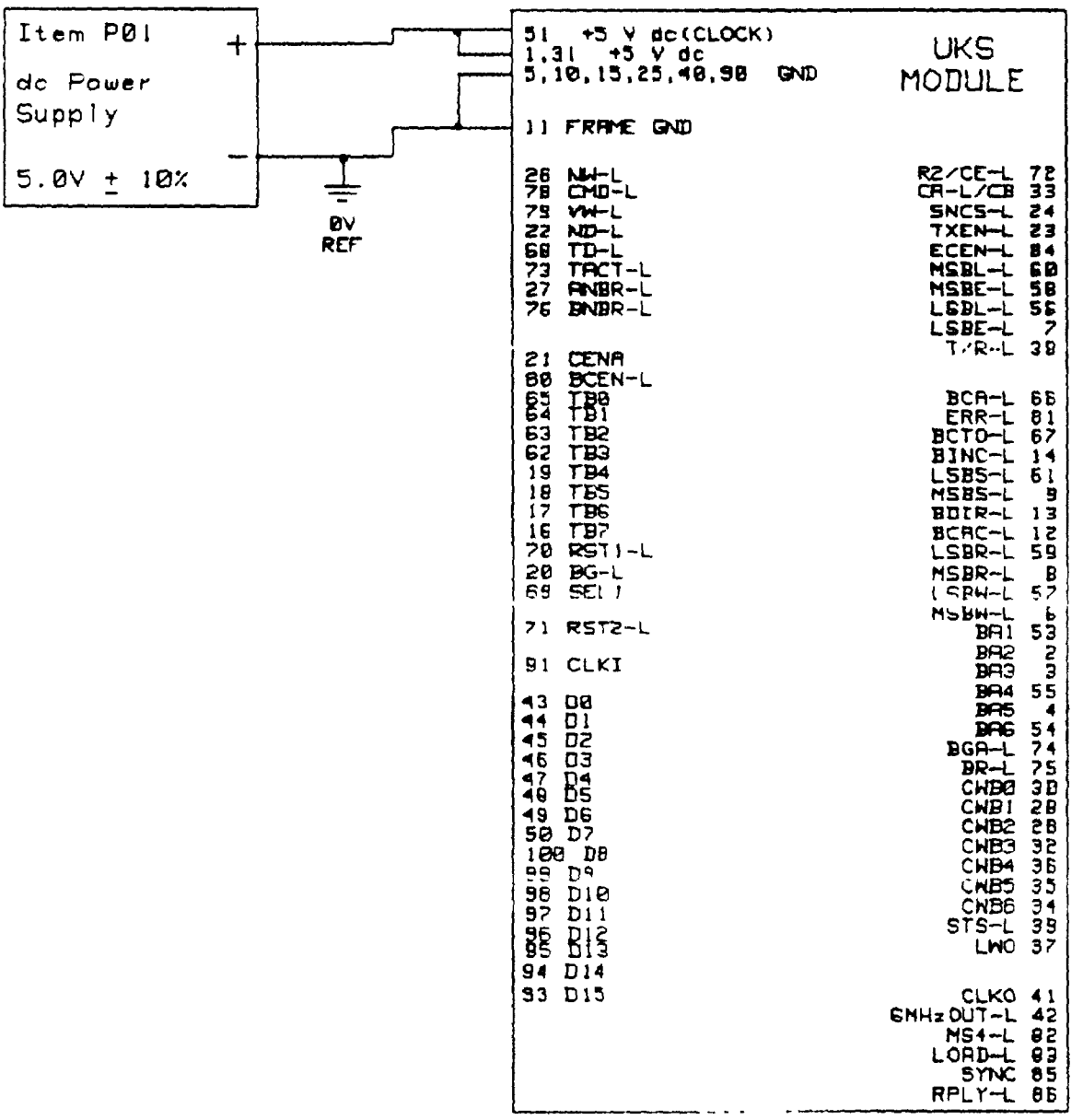


FIGURE 4. UKS vibration test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) Iozl = output low level leakage current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc.

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
TB0-TB7	-1.5	0.7	2.0	-0.4	20	100	40
BG-L	-1.5	0.7	2.0	-0.4	20	100	40
CENA	-1.5	0.7	2.0	-0.4	20	100	40
ND-L	-1.5	0.7	2.0	-0.4	20	100	40
NW-L	-1.5	0.7	2.0	-0.4	20	100	40
ANBR-L	-1.5	0.7	2.0	-0.8	-50	100	40
BNBR-L	-1.5	0.7	2.0	-0.8	-50	100	40
D8-D15	-1.5	0.7	2.0	-0.4	20	100	40
TD-L	-1.5	0.7	2.0	-0.4	20	100	40
RST2-L	-1.5	0.7	2.0	-0.8	-50	100	40
TACT-L	-1.5	0.7	2.0	-0.4	20	100	40
CLKI	-1.5	0.7	2.0	-0.4	20	100	40
RST1-L	-1.5	0.7	2.0	-0.4	20	100	40
BCEN-L	-1.5	0.7	2.0	-1.6	150	100	40

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D0-D7	-1.5	0.7	2.0	-0.2	20	100	40
CMD-L	-1.5	0.8	2.0	-0.4	25	1000	40
VW-L	-1.5	0.8	2.0	-0.4	50	1000	40
SEL1	-1.5	0.7	2.0	-1.6	100	100	40

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)
LSBS-L	2.4	0.4	-1.0	12.0	20	-20	-130
MSBS-L	2.4	0.4	-1.0	12.0	20	-20	-130
BDIR-L	2.4	0.4	-1.0	12.0	20	-20	-130
BCAC-L	2.4	0.4	-1.0	12.0	20	-20	-130
CA-L/CB	2.4	0.4	-3.0	12.0	20	-20	-225
SNCS-L	2.4	0.4	-3.0	12.0	20	-20	-225
TXEN-L	2.4	0.4	-3.0	12.0	20	-20	-225
T/R-L	2.4	0.4	-3.0	12.0	20	-20	-225
BGA-L	2.4	0.4	-3.0	12.0	20	-20	-225
BR-L	2.4	0.4	-3.0	12.0	20	-20	-225
BA1-BA6	2.4	0.4	-3.0	12.0	20	-20	-225
ECEN-L	2.4	0.5	-0.8	8.0	145	-750	-130
ERR-L	2.4	0.5	-0.8	8.0	100	-100	-130
MS4-L	2.4	0.5	-0.8	8.0	125	-350	-130

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)
LOAD-L	2.4	0.5	-0.8	8.0	205	-1950	-130
SYNC	2.4	0.5	-0.8	8.0	225	-1950	-130
RPLY-L	2.4	0.5	-0.8	8.0	285	-2950	-130
R2/CE-L	2.4	0.4	-0.4	4.0	NA	NA	-100
BCA-L	2.4	0.4	-0.4	4.0	NA	NA	-100
BCTO-L	2.4	0.4	-0.4	4.0	NA	NA	-100
CWB0-CWB6	2.4	0.4	-0.4	4.0	NA	NA	-100
STS-L	2.4	0.4	-0.4	4.0	NA	NA	-100
MSBL-L	2.4	0.4	-3.0	12.0	-100	-350	-225
MSBE-L	2.4	0.4	-3.0	12.0	-100	-350	-225
LSBL-L	2.4	0.4	-3.0	12.0	-100	-350	-225
LSBE-L	2.4	0.4	-3.0	12.0	-100	-350	-225
LSBR-L	2.4	0.4	-3.0	12.0	-100	-350	-225
MSBR-L	2.4	0.4	-3.0	12.0	-100	-350	-225

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl max (uA)	Ios max (mA)
LSBW-L	2.4	0.4	-3.0	12.0	-100	-350	-225
MSBW-L	2.4	0.4	-3.0	12.0	-100	-350	-225
LWO	2.4	0.4	-0.4	4.0	NA	NA	-100
6MHz-L	2.4	0.4	-3.0	8.0	NA	NA	-225
BINC-L	2.4	0.4	-0.4	1.6	NA	NA	-100
CLKO	4.1	0.5	-10.0	10.0	NA	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
1	tphl	CLKI (+) to BR-L (-)	-	-	-	220	220	220	nS	5,6
2	tplh	CLKI (+) to BR-L (+)	-	-	-	220	220	220	nS	5,6
3	tphl	CLKI (+) to BGA-L (-)	-	-	-	220	220	220	nS	5,6
4	tplh	CLKI (+) to BGA-L (+)	-	-	-	220	220	220	nS	5,6
5	tphl	CLKI (+) to MSBS-L (-) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
6	tplh	CLKI (+) to MSBS-L (+) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
7	tphl	CLKI (+) to LSBS-L (-) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
8	tplh	CLKI (+) to LSBS-L (+) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
9	tphl	CLKI (+) to BCAC-L (-) (BGA-L low)	-	-	-	220	220	220	nS	5,6

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
10	tplh	CLKI (+) to BCAC-L (+) (BGA-L low)	-	-	-	220	220	220	nS	5,6
11	tphl	CLKI (+) to MSBW-L (-) (RPLY-L low, BGA-L low)	-	-	-	220	220	220	nS	6
12	tplh	CLKI (+) to MSBW-L (+) (RPLY-L low, BGA-L low)	-	-	-	220	220	220	nS	6
13	tphl	CLKI (+) to LSBW-L (-) (RPLY-L low, BGA-L low)	-	-	-	220	220	220	nS	6
14	tplh	CLKI (+) to LSBW-L (+) (RPLY-L low, BGA-L low)	-	-	-	220	220	220	nS	6
15	tphl	CLKI (+) to MSBL-L (-) (LOAD-L low, BGA-L low)	-	-	-	220	220	220	nS	5
16	tplh	CLKI (+) to MSBL-L (+) (LOAD-L low, BGA-L low)	-	-	-	220	220	220	nS	5
17	tphl	CLKI (+) to LSBL-L (-) (LOAD-L low, BGA-L low)	-	-	-	220	220	220	nS	5

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
18	tplh	CLKI (+) to LSBL-L (+) (LOAD-L low, BGA-L low)	-	-	-	220	220	220	nS	5
19	tphl	CLKI (+) to MSBR-L (-) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
20	tplh	CLKI (+) to MSBR-L (+) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
21	tphl	CLKI (+) to LSBR-L (-) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
22	tplh	CLKI (+) to LSBR-L (+) (BGA-L low, BDIR-L high)	-	-	-	220	220	220	nS	5
23	tphl	CLKI (+) to MSBE-L (-) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
24	tplh	CLKI (+) to MSBE-L (+) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
25	tphl	CLKI (+) to LSBE-L (-) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
26	tplh	CLKI (+) to LSBE-L (+) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
27	tphl	CLKI (+) to BINC-L (-) (BGA-L low)	-	-	-	220	220	220	nS	5,6
28	tplh	CLKI (+) to BINC-L (+) (BGA-L low)	-	-	-	220	220	220	nS	5,6
29	tplh	CLKI (+) to BA1-BA6 (+) (Change occurs on each BINC-L (+))	-	-	-	220	220	220	nS	5,6
29	tphl	CLKI (+) to BA1-BA6 (-)	-	-	-	220	220	220	nS	5,6
	tphl	CLKI (+) to ERR-L (-) 3/	-	-	-	220	220	220	nS	
	tplh	CLKI (+) to ERR-L (+) 3/	-	-	-	220	220	220	nS	
29	tphl	CLKI (+) to MS4-L (-) 3/	-	-	-	220	220	220	nS	5,6
29	tplh	CLKI (+) to MS4-L (+) 3/	-	-	-	220	220	220	nS	5,6
29	tphl	CLKI (+) to LOAD-L (-) 3/	-	-	-	220	220	220	nS	5,6

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
29	tplh	CLKI (+) to LOAD-L (+) 3/	-	-	-	220	220	220	nS	5,6
30	tphl	CLKI (+) to ECEN-L (-) 3/	-	-	-	280	280	280	nS	5
31	tplh	CLKI (+) to ECEN-L (+) 3/	-	-	-	220	220	220	nS	6
	tphl	CLKI (+) to SYNC (-) 3/	-	-	-	220	220	220	nS	
	tplh	CLKI (+) to SYNC (+) 3/	-	-	-	220	220	220	nS	
31	tphl	CLKI (+) to RPLY-L (-) 3/	-	-	-	220	220	220	nS	6
	tplh	CLKI (+) to RPLY-L (+) 3/	-	-	-	220	220	220	nS	
32	tphl	CLKI (+) to CWB0-CWB6 (-) (Controller loading control word)	-	-	-	220	220	220	nS	5
32	tplh	CLKI (+) to CWB0-CWB6 (+) (BGA-L low)	-	-	-	220	220	220	nS	5
32	tphl	CLKI (+) to CA-L/CB (-) (Control word bit 7)	-	-	-	220	220	220	nS	5

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
32	tplh	CLKI (+) to CA-L/CB (+) (Control word bit 7)	-	-	-	220	220	220	nS	5
	tphl	CLKI (+) to T/R-L (-) (Changes during command word load sequence)	-	-	-	220	220	220	nS	
	tplh	CLKI (+) to T/R-L (+) (Changes during command word load sequence)	-	-	-	220	220	220	nS	
33	tphl	CLKI (+) to SNCS-L (-) (Changes during command data status word transfer)	-	-	-	220	220	220	nS	5
	tplh	CLKI (+) to SNCS-L (+) (Changes during command data status word transfer)	-	-	-	220	220	220	nS	

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
34	tphl	CLKI (+) to TXEN-L (-) (Goes low at completion of command word load cycle)	-	-	-	220	220	220	nS	5
35	tplh	TACT-L (+) to TXEN-L (+) (Goes low at completion of word load cycle)	-	-	-	220	220	220	nS	6
36	tphl	CLKI (+) to STS-L (-) (Active during status word check if error exists)	-	-	-	280	280	280	nS	6
37	tplh	CLKI (+) to LWO (+) 4/	-	-	-	220	220	220	nS	6
38	tphl	CLKI (+) to LWO (-) 4/	-	-	-	220	220	220	nS	6
	tphl	CLKI (+) to BCTO-L (-) 5/	-	-	-	220	220	220	nS	
	tphl	CLKI (+) to R2/CE-L (-) 6/	-	-	-	220	220	220	nS	
	tphl	RST2-L (-) to R2/CE-L (-) 6/	-	-	-	40	40	40	nS	

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
39	tphl	CLKI (+) to MSBS-L (-) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
40	tplh	CLKI (+) to MSBS-L (+) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
41	tphl	CLKI (+) to LSBS-L (-) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
42	tplh	CLKI (+) to LSBS-L (+) (BGA-L low, BDIR-L low)	-	-	-	220	220	220	nS	6
	tphl	CLKI (+) to BDIR-L (-)	-	-	-	220	220	220	nS	6
	tplh	CLKI (+) to BDIR-L (-)	-	-	-	220	220	220	nS	6
43	tpwl	BR-L 7/	9	9	9	10	10	10	6MHz-L Cycles	5,6
44	tpwl	BGA-L 8/	8	8	8	8	8	8	6MHz-L Cycles	5,6
45	tpwl	MSBS-L (BGA-L low, BDIR-L high)	3	3	3	3	3	3	6MHz-L Cycles	5
46	tpwl	LSBS-L (BGA-L low, BDIR-L high)	3	3	3	3	3	3	6MHz-L Cycles	5

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
47	tpwl	MSBS-L (BGA-L low, BDIR-L low)	1	1	1	1	1	1	6MHz-L Cycles	6
48	tpwl	LSBS-L (BGA-L low, BDIR-L low)	1	1	1	1	1	1	6MHz-L Cycles	6
49	tpwl	BCAC-L (BGA-L low)	3	3	3	3	3	3	6MHz-L Cycles	5,6
50	tpwl	MSBW-L (RPLY-L low, BGA-L low)	1	1	1	1	1	1	6MHz-L Cycles	6
51	tpwl	LSBW-L (RPLY-L low, BGA-L low)	1	1	1	1	1	1	6MHz-L Cycles	6
52	tpwl	MSBL-L (LOAD-L low, BGA-L low)	1	1	1	1	1	1	6MHz-L Cycles	5
53	tpwl	LSBL-L (LOAD-L low, BGA-L low)	1	1	1	1	1	1	6MHz-L Cycles	5
54	tpwl	MSBR-L (BGA-L low, BDIR-L high)	3	3	3	3	3	3	6MHz-L Cycles	5
55	tpwl	LSBR-L (BGA-L low, BDIR-L high)	3	3	3	3	3	3	6MHz-L Cycles	5
56	tpwl	MSBE-L (BGA-L low, BDIR-L low)	3	3	3	3	3	3	6MHz-L Cycles	6

- See footnotes at end of table.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter 2/	Minimum limit 1/			Maximum limit 1/			Unit	Fig
			-55°	25°	85°	-55°	25°	85°		
57	tpwl	LSBE-L (BGA-L low, BDIR-L low)	3	3	3	3	3	3	6MHz-L Cycles	6
58	tpwl	BINC-L (BGA-L low)	1	1	1	1	1	1	6MHz-L Cycles	5,6
59	tpwl	STS-L (BGA-L low)	1	1	1	1	1	1	6MHz-L Cycles	6

- See footnotes on next page.

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1/ The limits for class I at 0, 25, and 60°C are the same as class II limits at -55, 25, and 85°C.

2/ (+) following a signal name indicates a positive going pulse edge and (-) following a signal name indicates a negative going pulse edge.

3/ These six outputs (ERR-L, MS4-L, LOAD-L, ECEN-L, SYNC, RPLY-L) change in response to conditions defined by input signals to allow the sequencer to change state. Time periods are dependent on bus activity.

4/ Goes active at completion of last word transfer; cleared by next transfer.

5/ Goes active if no response time is exceeded on 1553 bus. Reset by RST1-L or CENA.

6/ Pulse width determined by input pulse width.

7/ BG-L low, 8 BIT MODE. Subtract FIVE for 16 bit mode.

8/ BR-L, BG-L low, 8 BIT MODE. Subtract FOUR for 16 bit mode.

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6.4 Module function. The UKS provides the necessary control signals to implement a MIL-STD-1553 bus controller when used with the UKT bus interface SEM. The bus controller may also be configured as a combination Remote Terminal/Bus Controller (RT/BC) by the addition of a UKU module. Module functional waveforms are defined on figures 5 through 7.

6.4.1 Operation.

(a) The bus controller requires a minimum number of control signals from the subsystem it is attached to. All data transfer signals are generated by the bus controller as required to complete message transfer operations. All front end control signals needed by the UKT are supported in order to configure the SEM set with no additional devices required.

(b) BCEN-L, RST1-L, BG-L, and CENA are the control inputs required from the subsystem for controller operation.

(c) BCEN-L is the master select line for the controller. When high, the controller is idle and all three state lines are inactive. The sequencer is not selected and will not accept the cycle enable command.

(d) RST1-L is the hardware reset input and should be connected to the subsystem reset signal or power-on reset circuitry.

(e) Cycle enable is an active high input. When held low, the controller and R2/CE-L signal are in reset. When cycle enable goes high, the bus controller will begin a message transfer sequence, loading the control word, command word(s) and data transfer as required.

(f) BG-L is the handshake signal issued by the subsystem in response to controller BR-L (Bus Request) output, indicating data transfer is required. When BG-L goes active low, the controller responds with BGA-L (Bus Grant Acknowledge) and complete data transfer cycle.

(g) The controller sequence will perform the appropriate data transfer operations in response to the control word loaded at the beginning of the message transfer sequence. Upon completion of the message transfer sequence, the controller will enter an idle state, requiring cycle enable to go low to reset the controller.

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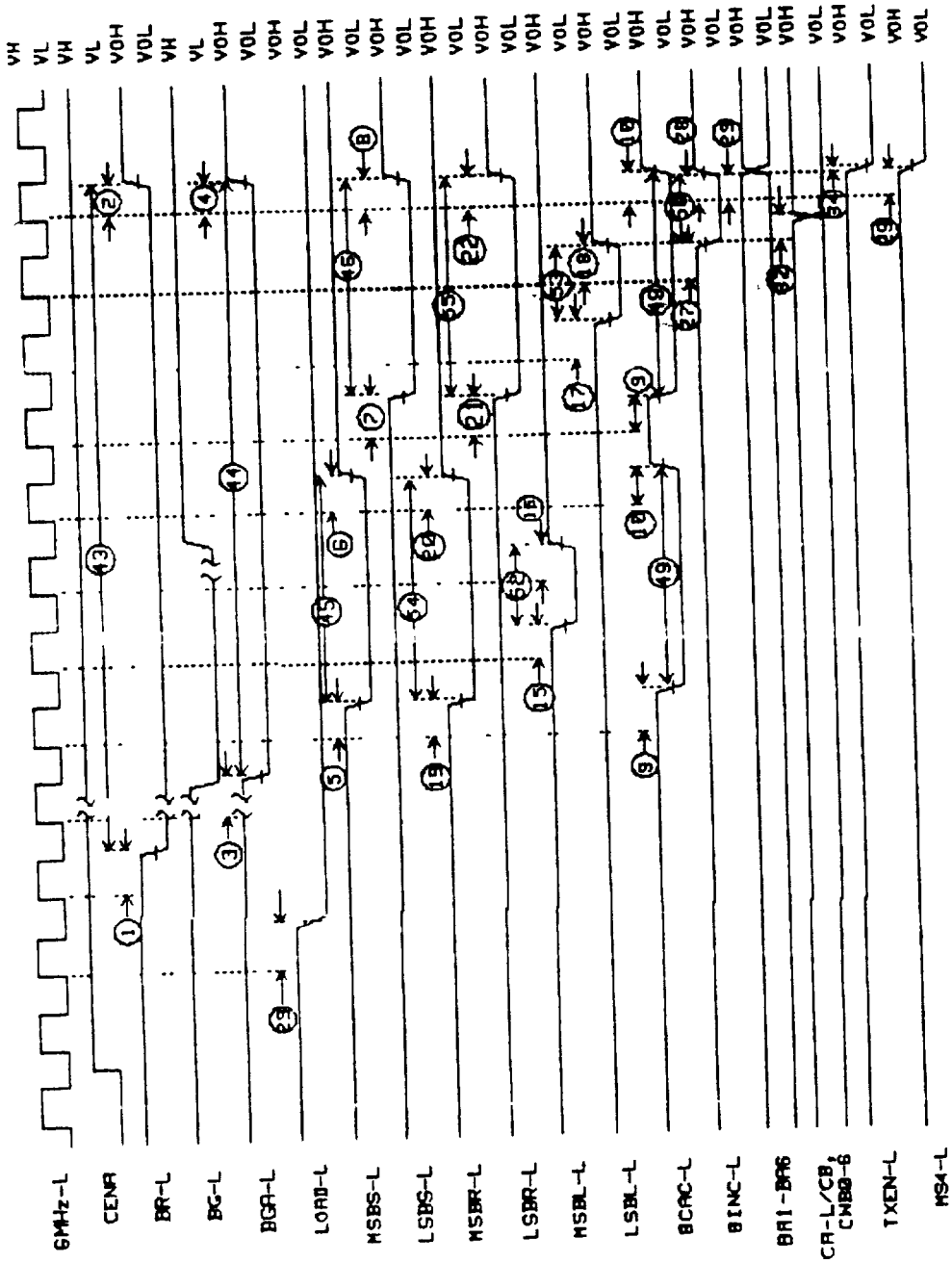


FIGURE 5a. RT receive command (write to front end), 8 bit mode control word load cycle.

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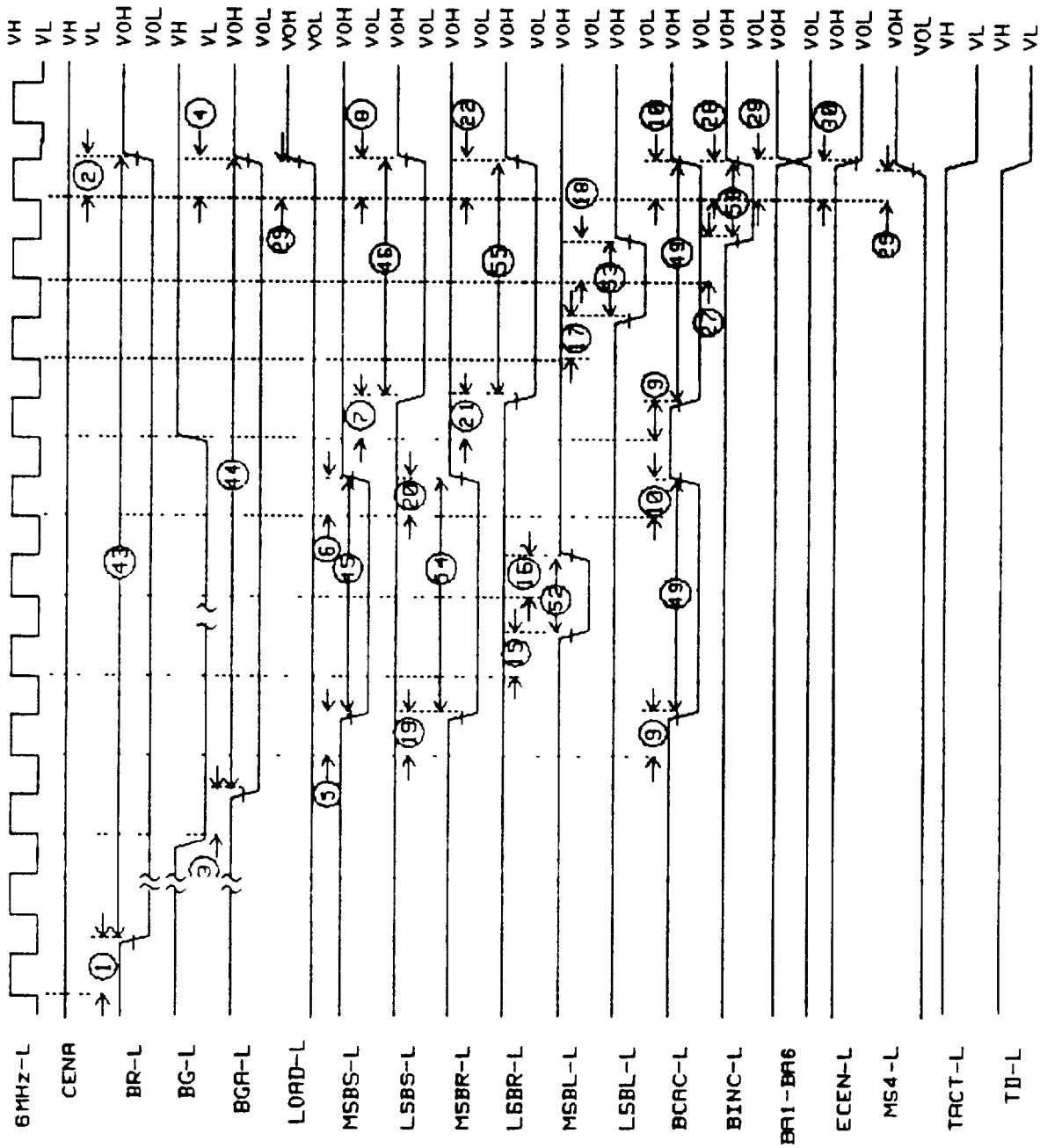


FIGURE 5b. RT receive command (write to front end), 8 bit mode data word transfer.

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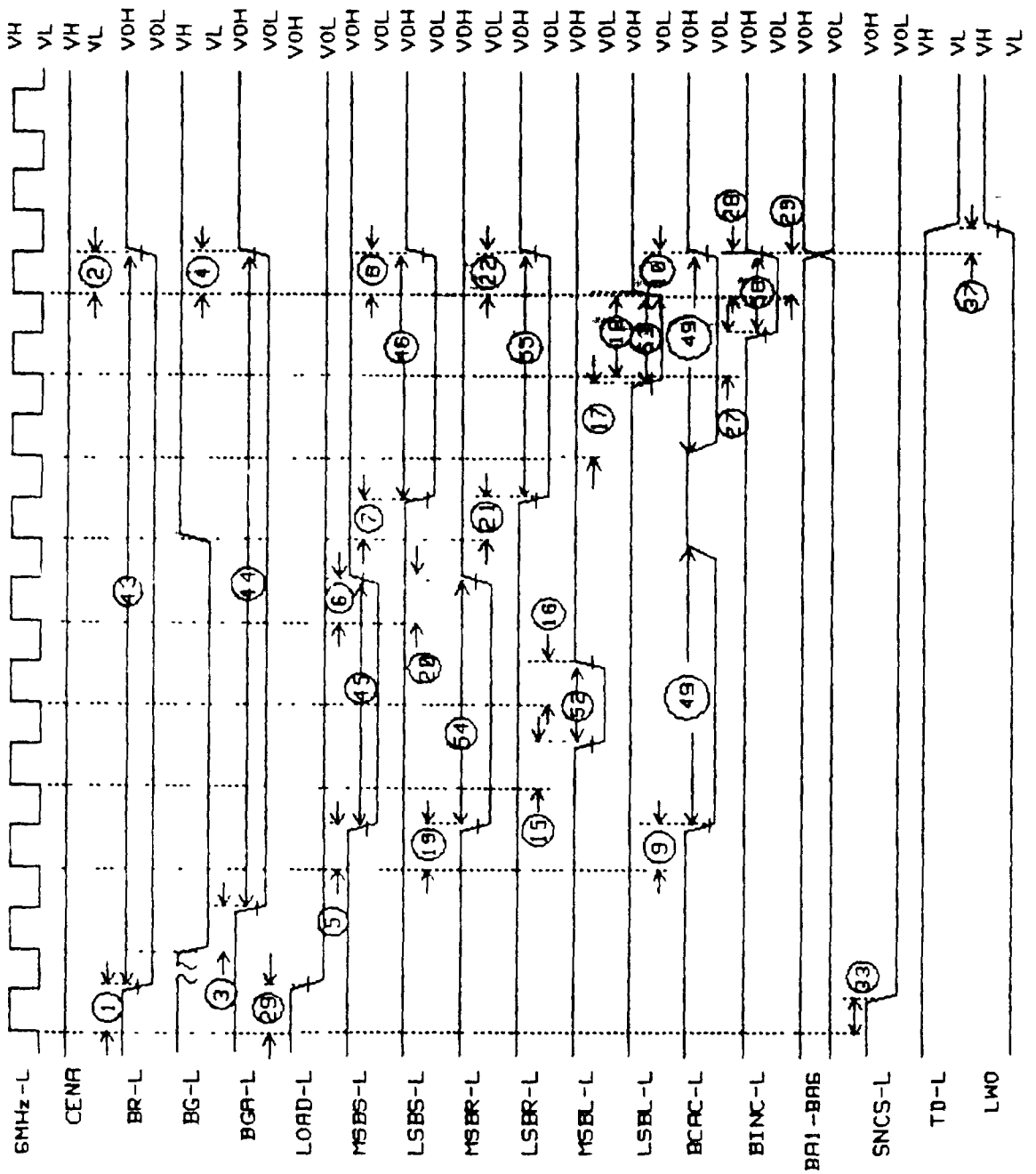


FIGURE 5c. RT receive command (write to front end), 8 bit mode data word transfer.

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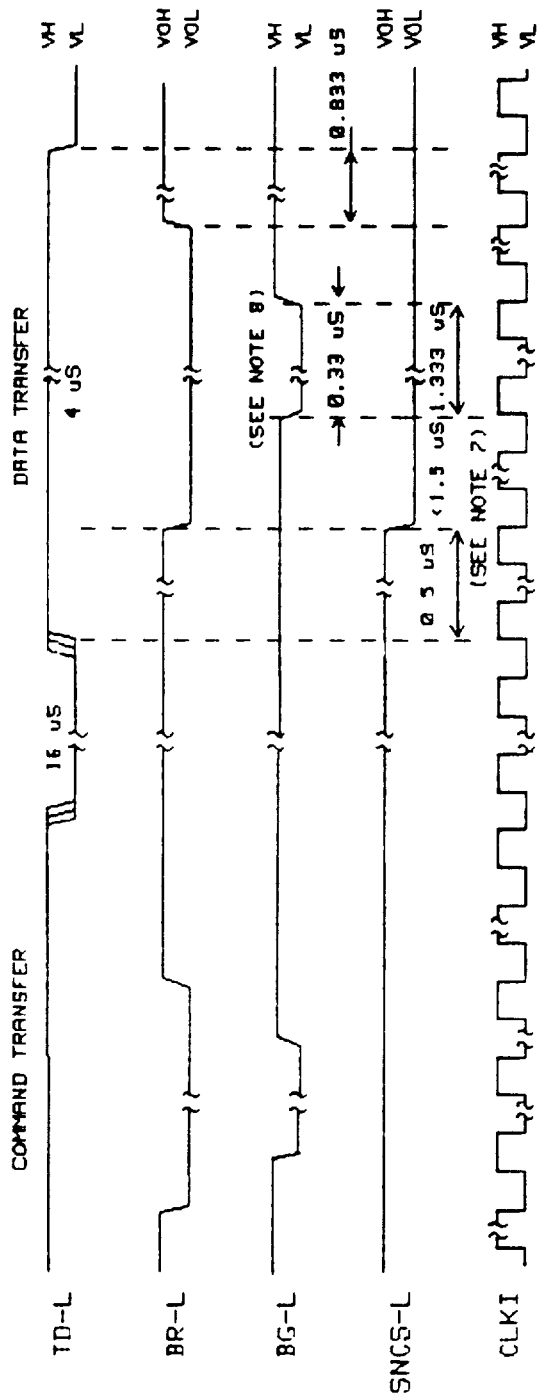


FIGURE 5d. RT receive command (write to front end), interword gap/front transmit detail.

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NOTES:

1. Bus controller transfer sequence.

(a) Subsystem raises cycle enable input to controller.

(b) Bus controller loads control word into control register and enters appropriate message transfer format.

2. Receive command.

(a) If T/R bit in command word is low, bus controller issues a receive command to appropriate RT.

(b) As command is loaded into front end, bus controller word counter is initialized to provide LWO signal after last data word is loaded into front end.

(c) Bus controller monitors TD-L to determine when front end has completed each word transmission and loads next data word during parity and sync transmit time (4 uS).

(d) If last word is active, bus controller performs loopback of last transmitted word and enters wait loop for status word from RT. BCTO-L counter is initialized to provide BCTO-L signal if RT does not respond within programmed response time period.

(e) ECEN-L is turned off when TD-L goes high and LWO is active.

(f) TXEN-L is turned off when TACT-L goes high following last data word.

3. Broadcast command.

(a) If broadcast bit is set, the bus controller will follow 2(a)-2(f) above with the following exceptions:

(1) BCTO-L will not be initialized.

(2) Controller will not enter wait for status as none is expected for a broadcast command.

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NOTES: - Continued.

4. RT to RT transfer.

(a) Bus controller will issue receive and transmit commands to RTs.

(b) Controller will initialize BCTO-L counter and enter wait for status from each RT.

5. Mode code.

(a) Bus controller will follow steps required to transfer appropriate command and data.

6. Last word loopback.

(a) For all message transfers to the front end, the controller will perform a last word loopback to subsystem memory. This function is accomplished using the front end receiver circuitry and ensures that the data was properly transmitted onto the 1553 bus. Data is stored into subsystem memory at the next logical address following data transfer.

(b) After loopback, the controller then proceeds to its next sequence, entering receiver wait loop for data or status or completing command if no receive data is required.

7. Time BR-L (high to low) to BG-L (high to low) is controlled by subsystem and must be less than 1.5 uS 8 bit mode and 2.167 uS in 16 bit mode during collective word transfers in order to avoid gaps on the 1553 bus.

8. Duration of BG-L low should be at least 2 clock cycles (0.333 uS) and should go high prior to BINC-L (low to high).

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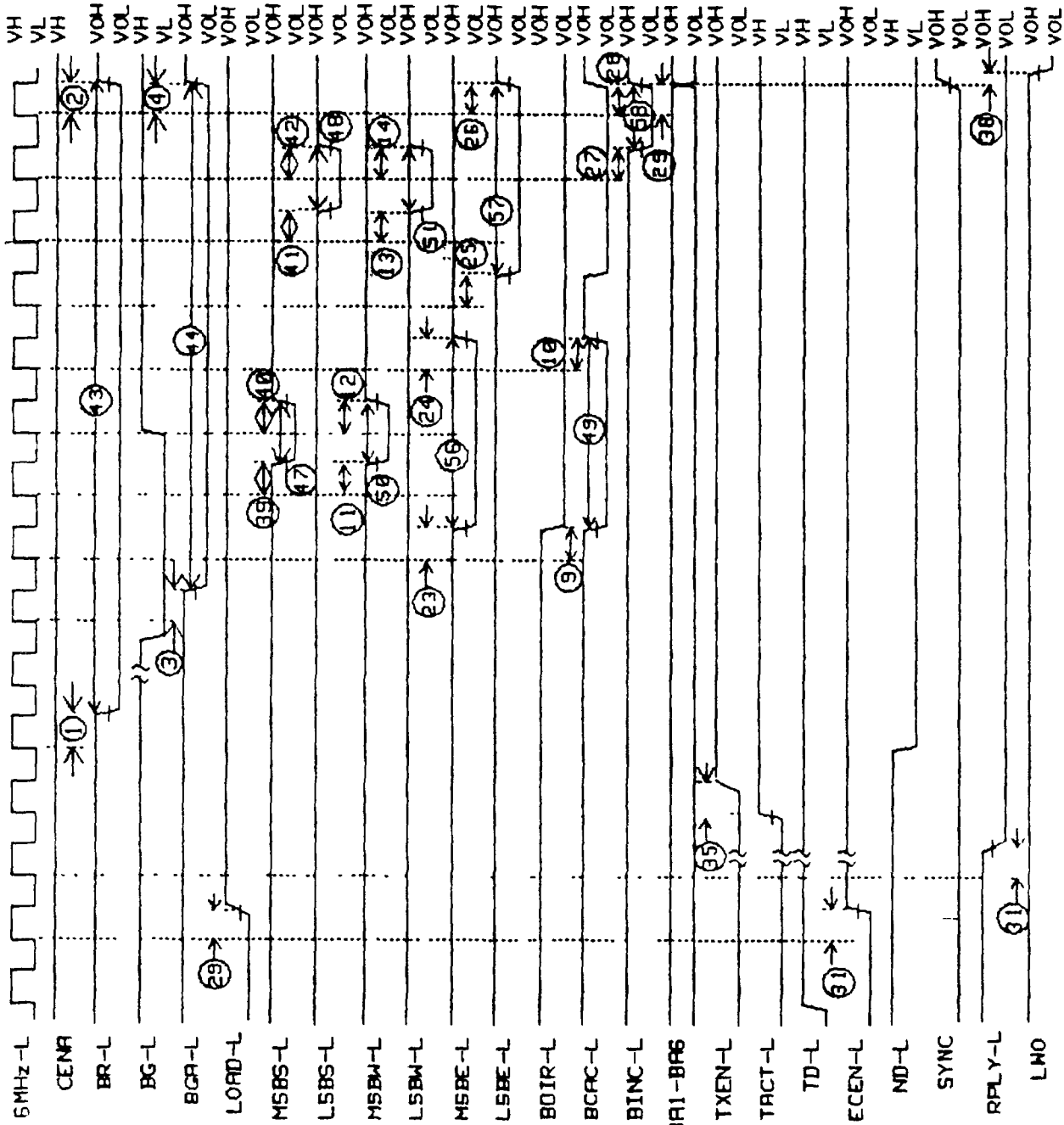


FIGURE 6a. RT transmit (write to subsystem), 8 bit mode, last word wraparound.

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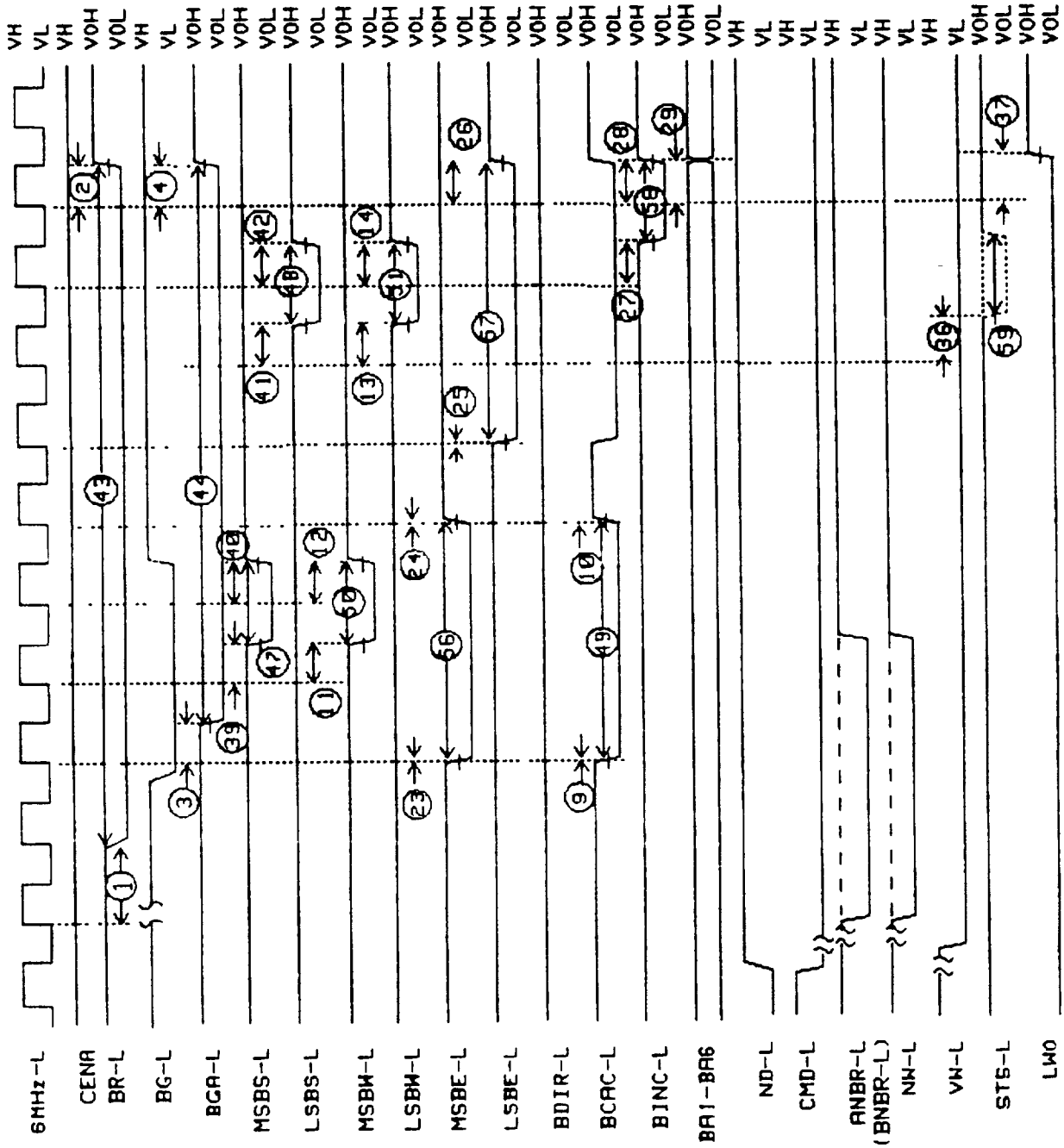


FIGURE 6b. RT transmit (write to subsystem), 8 bit mode, bus controller transfer to subsystem command/status word.

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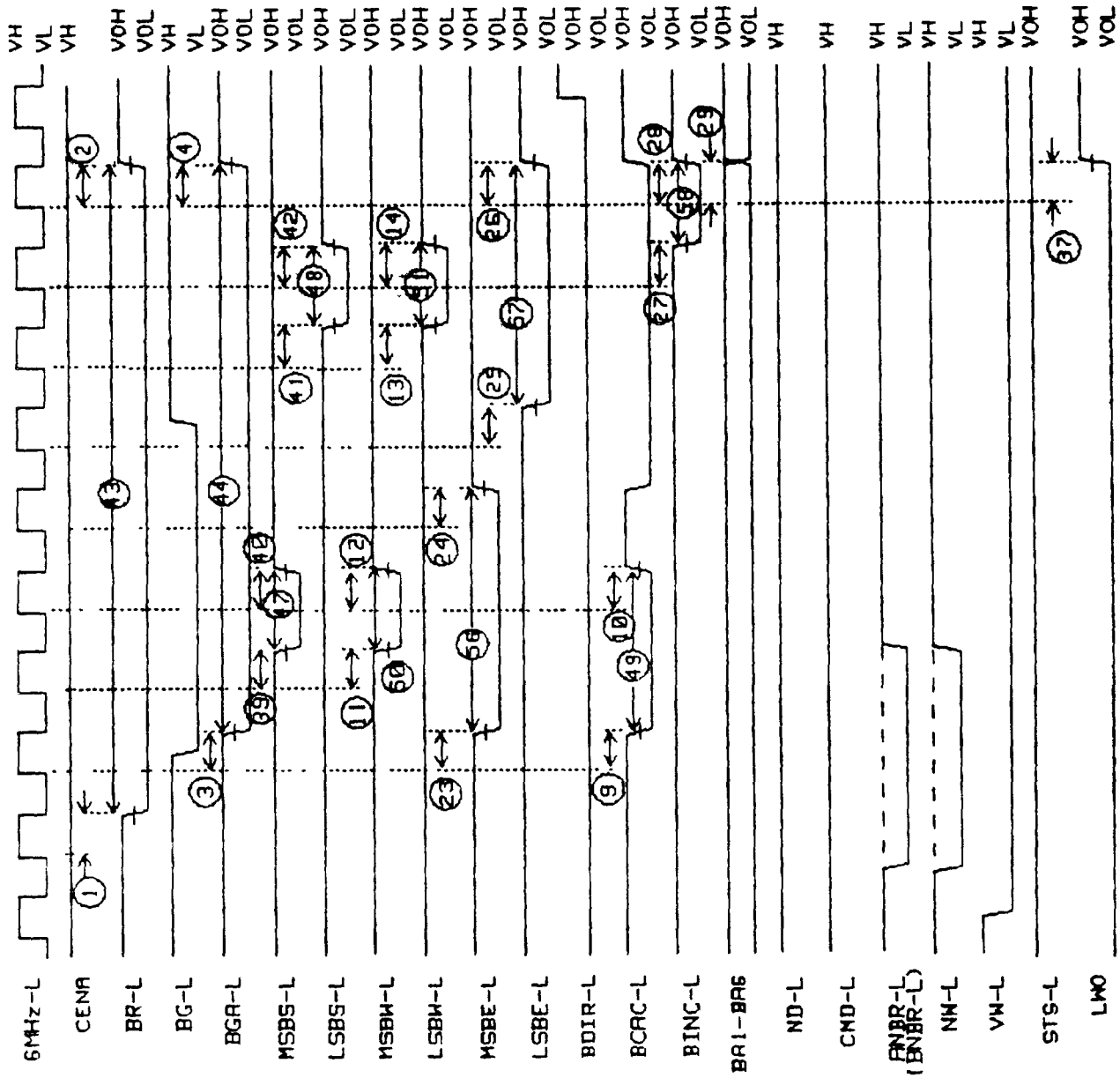


FIGURE 6c. RT transmit (write to subsystem), 8 bit mode, bus transfer to subsystem data word.

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NOTES:

1. Bus controller transfer sequence.

(a) Subsystem raises cycle enable to controller.

(b) Bus controller loads control word into control register and enters appropriate transfer sequence.

2. Transmit command.

(a) If T/R-L bit is high, bus controller issues transmit command to 1553 bus.

(b) After front end transmits command, controller performs loopback of last word and waits for status.

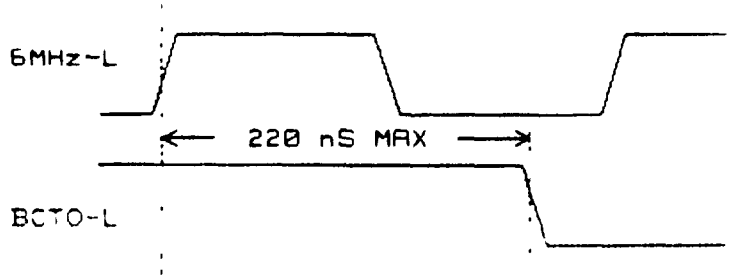
(c) If status word is incorrect, the controller will issue STS-L pulse to subsystem and continue until aborted.

(d) After status is received, controller will begin servicing data transfers until last word is received, and will enter an idle state.

(e) If data received is not valid or has the wrong SYNC type, the sequencer will trigger ERR-L, indicating a failure during message transfer.

(f) During idle, controller will continue to respond to front end requests for transfer, capturing any data which may be present on the bus.

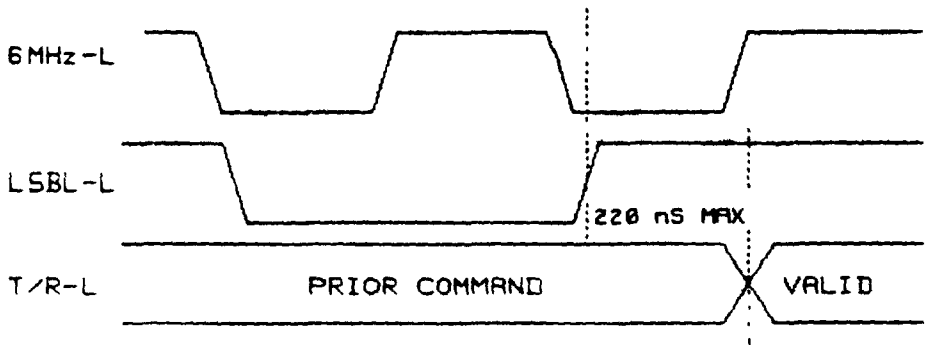
(g) BCTO-L is used to indicate that a remote terminal has not responded within the time set by the TB0-TB7 inputs. The time determined by these inputs are in 166 nS increments. The BCTO-L signal is generated by the terminal count.



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NOTES: - Continued.

(h) T/R-L becomes valid, as shown below, after the command word is written into the front end.



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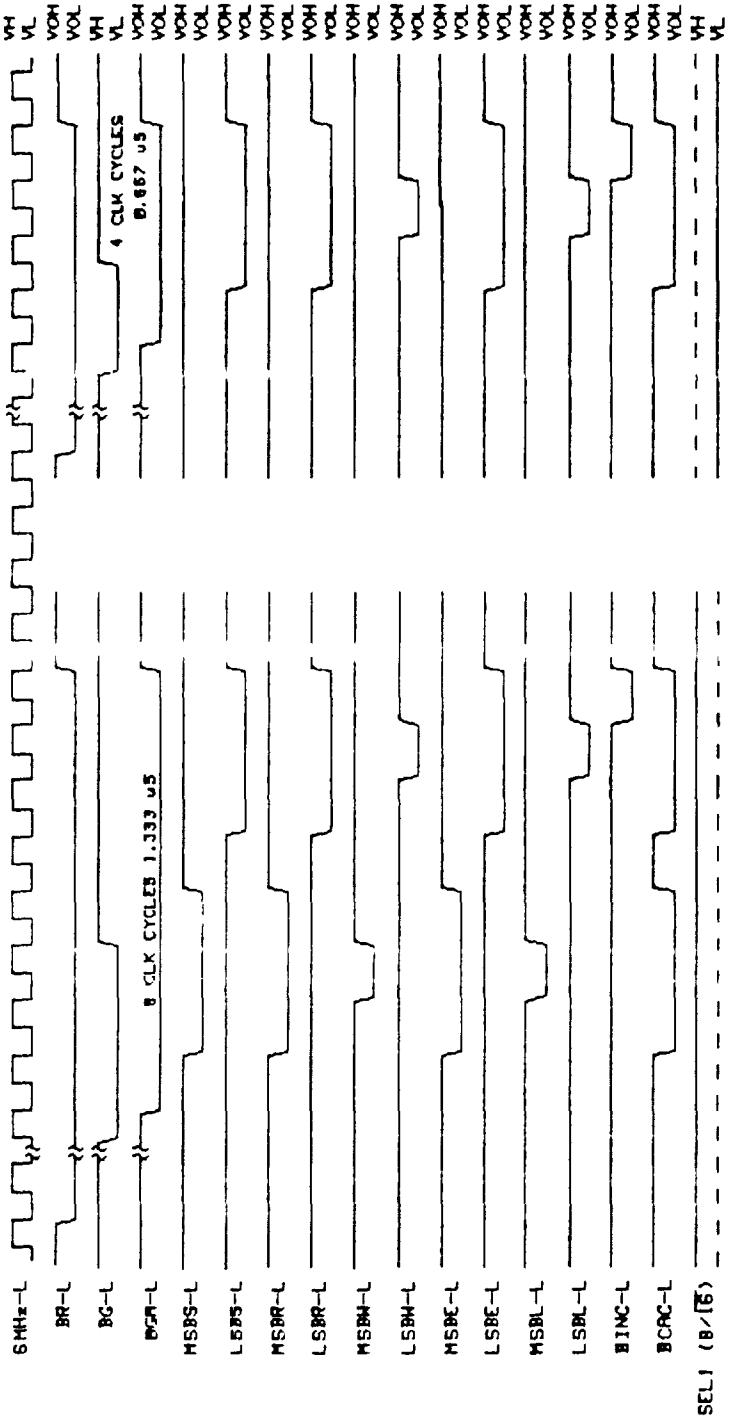


FIGURE 7. Bus cycle comparison - 8 bit vs 16 bit mode.

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NOTE:

1. This diagram illustrates the difference in cycle operation between 8 bit and 16 bit operation. The controller requires four additional 6 MHz clock cycles to complete a data bus transfer in 8 bit mode. This has the effect of reducing subsystem response time during consecutive data transfers. In 8 bit mode, the controller will transfer the most significant byte followed by the least significant byte. In 16 bit mode, the controller issues LSB signals to transfer the 16 bit word. All MSB signals are inactive during 16 bit mode.

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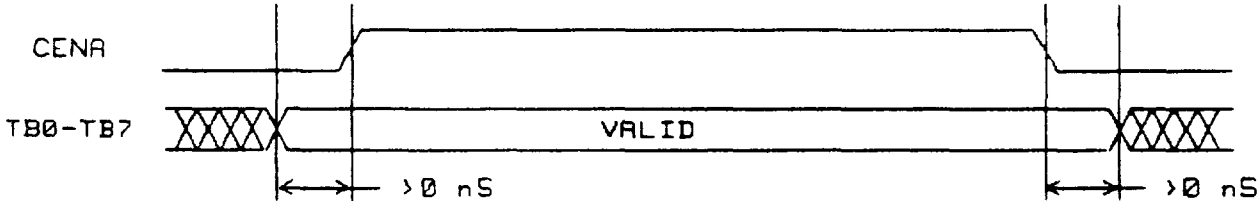
6.4.2 Input timing. Subsystem/control inputs.

6.4.2.1 Static inputs. The following inputs should be put into the correct state and held for the entire message time. In most cases, these inputs are "hardwired" high or low.

6.4.2.1.1 SEL1. SEL1 is high for 8 bit mode, and low for 16 bit mode.

6.4.2.1.2 BCEN-L. A low level signal enables sequencer and all three-state outputs. If the system is configured as RT/BC, this input shall be driven by the same signal that generates RTDIS-L to the UKU module. This will prevent bus contention between the two modules.

6.4.2.1.3 TB0-TB7. Signals TB0-TB7 are no response timer inputs. They should be held stable for complete MSG transfer. They may be changed prior to CENA active if variable delays are required. (See figure 8.)



NOTE:

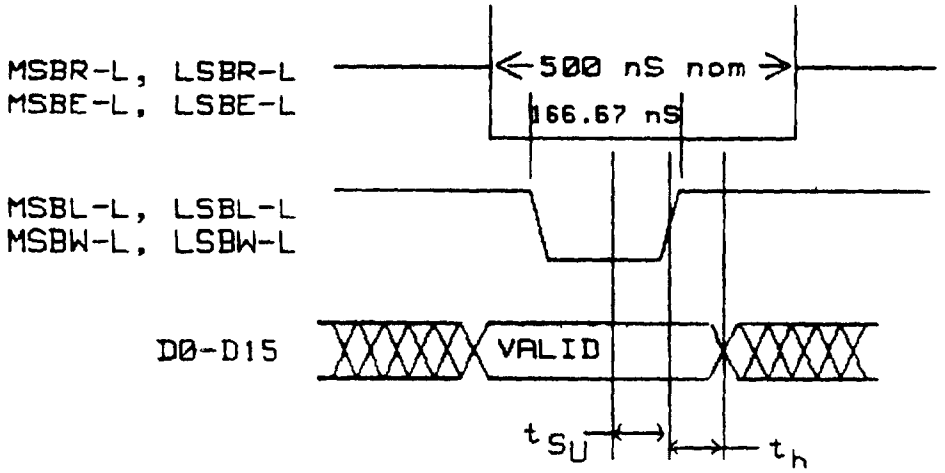
1. tsu > 0 nS, th > 0 nS.

FIGURE 8. TB0-TB7 timing characteristics.

6.4.2.2 Dynamic inputs.

6.4.2.2.1 D0-D15. Input data must be valid during MSBL-L, LSBL-L, MSBE-L, LSBE-L, MSBR-L, LSBR-L, MSBW-L, LSBW-L, MSBS-L, or LSBS-L. These signals control the reading and writing of data in the front end registers and subsystem memory. As inputs to the module, data is required during certain MSBL-L or LSBL-L pulses to allow CTRL, CMD, and status decoding of the data on the bus. (See figure 9.)

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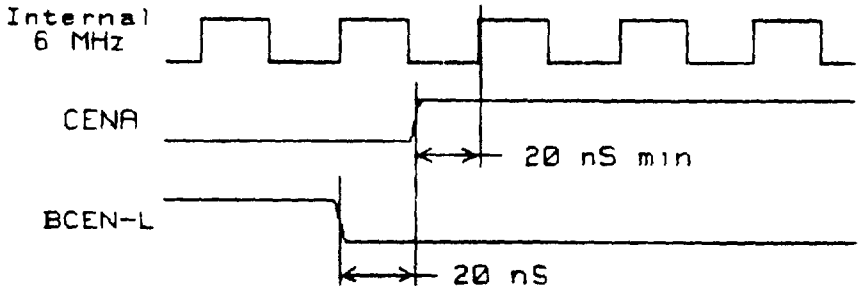


NOTE:

1. tsu > 40 nS, th > 40 nS.

FIGURE 9. D0-D15 timing characteristics.

6.4.2.2.2 CENA. CENA is the asynchronous active high signal. It is held high for complete message transfer. It is latched by rising edge of internal 6 MHz clock. (See figure 10.)



NOTE:

1. BCEN-L - Output enable should be low prior to CENA high.

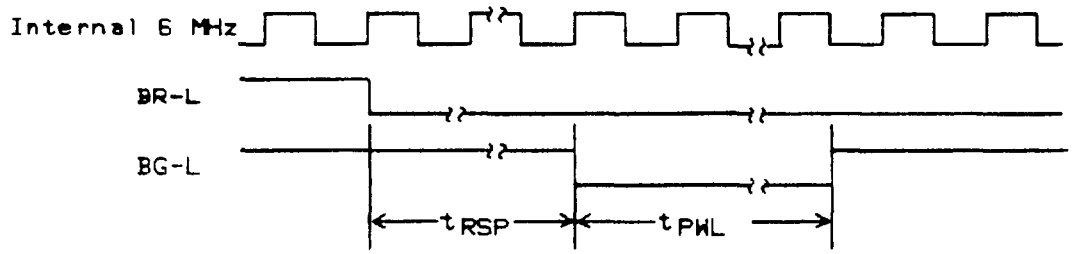
FIGURE 10. CENA timing characteristics.

6.4.2.2.3 RST1-L. RST1-L is the master reset input. It is an asynchronous input. Minimum low level pulse width is 200 nS.

6.4.2.2.4 RST2-L. RST2-1 is the reset input used to generate R2/CE-L which resets front end module. It is an asynchronous input. Minimum low level pulse width is 200 nS.

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6.4.2.2.5 BG-L. Bus Grant input used to enable data transfer on parallel bus in response to BR-L output. tRSP must be met during 1553 bus data transfer to avoid gaps. During CTRL word load, tRSP is not critical. (See figure 11.)



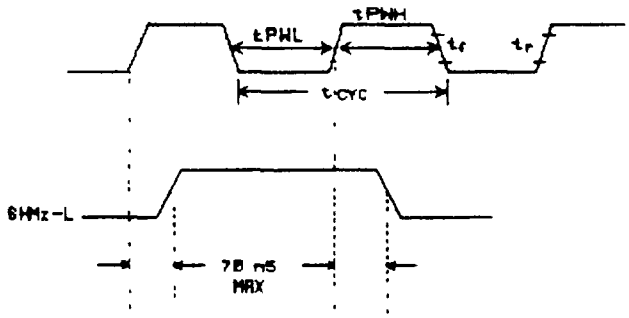
NOTES:

1. tRSP < 1.5 uS for 8 bit system, tRSP < 2.167 uS for 16 bit system.

2. 200 nS < tPNL < 2 uS.

FIGURE 11. BG-L timing characteristics.

6.4.2.2.6 12 MHz CLKI. CLKI is a 12 MHz TTL level square wave input. (See figure 12.)



NOTES:

1. tr = tf = 10 nS nominal.

2. tCYC = 83.333 nS nominal.

3. tPWL = 41.667 nS nominal.

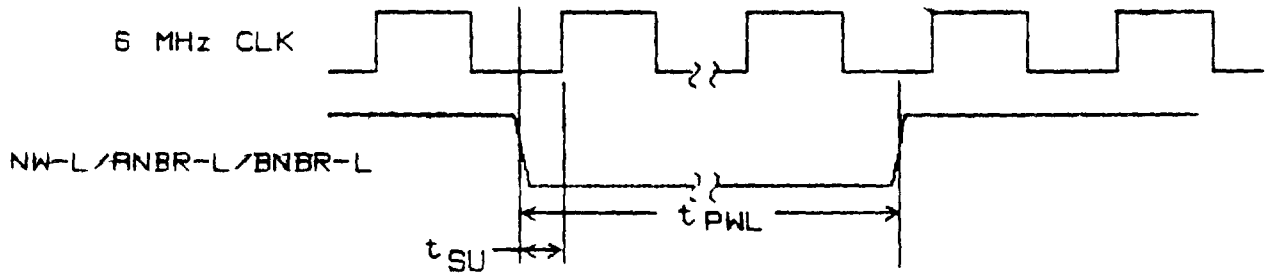
4. tPWH = 41.667 nS nominal.

FIGURE 12. 12 MHz CLKI timing characteristics.

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6.4.2.3 Input signals. Front end CTRL/response inputs.

6.4.2.3.1 NW-L/ANBR-L/BNBR-L. Inputs used to initiate front end read cycle by the controller. (See figure 13.)

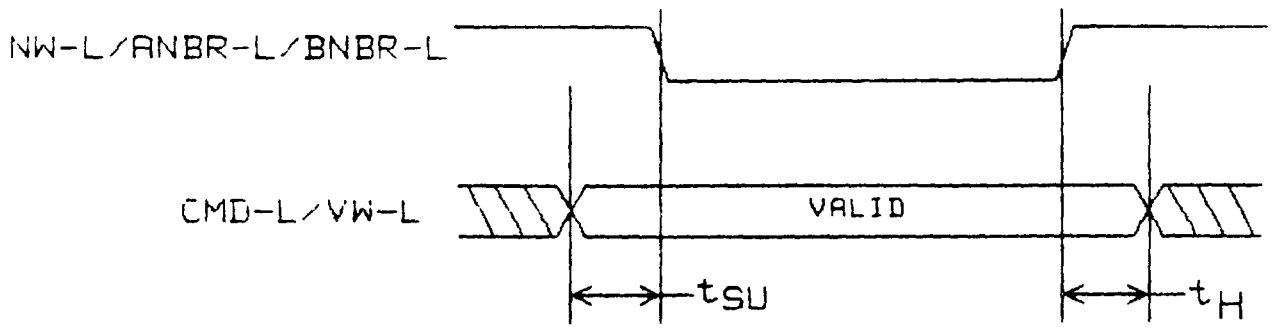


NOTES:

1. tsu > 40 nS.
2. 1 clock cycle < tPWI < 4 clock cycle.

FIGURE 13. NW-L/ANBR-L/BNBR-L timing characteristics.

6.4.2.3.2 CMD-L/VW-L. Inputs required valid during NW-L/ANBR-L/BNBR-L to allow sequencer to perform transfer and error checking. (See figure 14.)



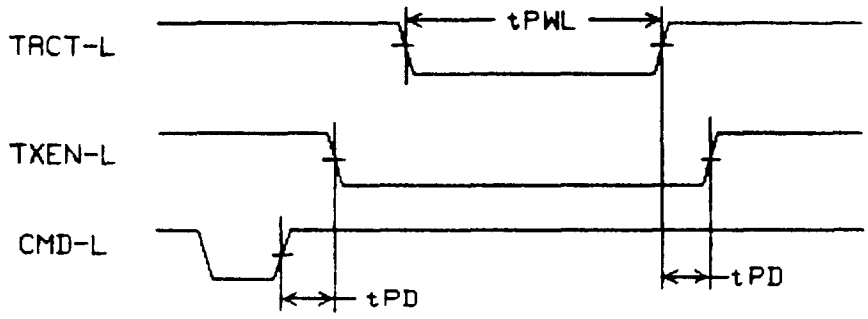
NOTE:

1. tsu > 130 nS, th > 300 nS.

FIGURE 14. CMD-L/VW-L timing characteristics.

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6.4.2.3.3 TACT-L. Input signal used to disable TXEN-L at completion of transmit. Active during data transmit on 1553 bus. (See figure 15.)

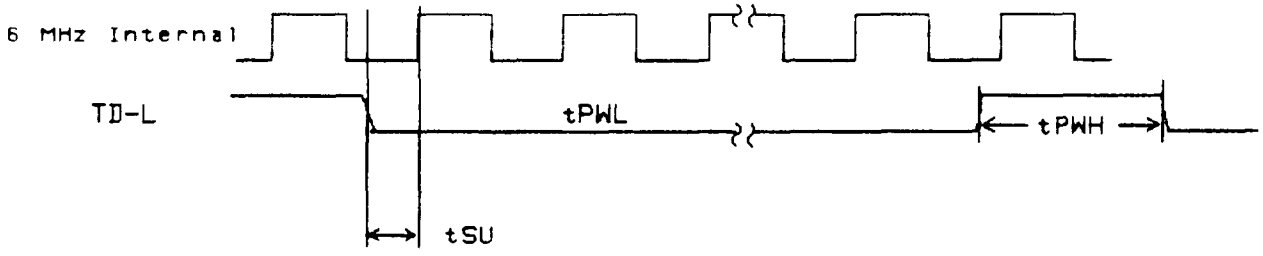


NOTES:

1. tPWL > 110 nS.
2. tPD < 120 nS.

FIGURE 15. TACT-L timing characteristics.

6.4.2.3.4 TD-L. Transmit data pulse active during transfer of 16 bit data word. Triggers sequencer to request additional data word(s) if required. (See figure 16.)



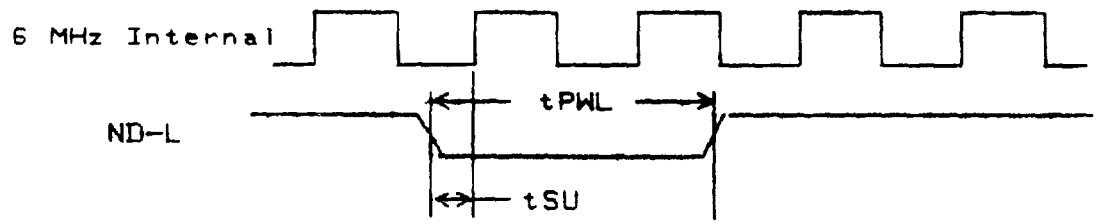
NOTES:

1. tPWL = 16 clocks.
2. tPWH = 4 clocks.
3. tsu > 30 nS.

FIGURE 16. TD-L timing characteristics.

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6.4.2.3.5 ND-L. No data input signal indicates 1553 bus is not active, triggers sequencer to provide wraparound of data during transmit, enables no response timer during controller receive mode. (See figure 17.)



NOTES:

1. tsu > 30 nS.
2. tPWL > 2 clocks.

FIGURE 17. ND-L timing characteristics.

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6.4.3 Functional description of I/O pins.

Functional designation	Pin number	I/O	Functional description
LSBS-L	61	O	Subsystem chip select output (active low) active during LSB transfer in 8 bit mode. Also active as subsystem CS-L during 16 bit mode.
MSBS-L	9	O	Chip select (active low) output active during MSB transfer in 8 bit mode. Used as chip select for subsystem memory.
BDIR-L	13	O	Bus direction (active low) output active during bus transfer, low for subsystem write, high for subsystem read by BC.
BCAC-L	12	O	BC access line (active low). Active whenever bus controller is transferring data on the parallel bus.
CA-L/CB	33	O	Channel A/B select line. Low selects channel A.
SNCS-L	24	O	Sync select output (active low). Low indicates command or status word.
TXEN-L	23	O	Transmitter enable output (active low). Enables front end transmitter section.
MSBL-L	60	O	Front end load (active low) signal active during MSB transfer in 8 bit mode.
MSBE-L	58	O	Front end enable (active low) signal active during MSB transfer in 8 bit mode. Not active in 16 bit mode.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
LSBL-L	56	O	Front end load (active low) signal active during transfer of LSB in 8 bit mode. Active as front end load (active low) during 16 bit mode. Loads data into transmitter buffer.
LSBE-L	7	O	Front end enable (active low) signal active during LSB transfer in 8 bit mode. Also used as data enable (active low) during 16 bit mode, enables front end receiver buffer onto parallel bus.
T/R-L	38	O	Transmit/receive signal indicates current command type. Receive is active low.
LSBR-L	59	O	Subsystem read (active low) signal active during LSB transfer in 8 bit mode. Also active as read (active low) during 16 bit mode. Used to read subsystem memory.
MSBR-L	8	O	Subsystem read (active low) signal active during MSB transfer in 8 bit mode. Used to read subsystem memory.
LSBW-L	57	O	Subsystem write (active low) signal active during transfer of LSB during 8 bit mode. Also operates as subsystem write (active) during 16 bit mode. Used to write data to subsystem memory.
MSBW-L	6	O	Subsystem write (active low) signal active during MSB transfer when in 8 bit mode.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
BA1	53	O	Least significant bus address line.
BA2	2	O	Bus address counter output.
BA3	3	O	Bus address counter output.
BA4	55	O	Bus address counter output.
BA5	4	O	Bus address counter output.
BA6	54	O	Most significant bus address line.
BGA-L	74	O	Bus grant acknowledge output (active low) indicates controller has received BG-L input and is engaged in a data transfer operation on the parallel bus.
BR-L	75	O	Bus request output (active low) issued by controller when data transfer is required, indicating bus access is required.
ECEN-L	84	O	Encoder enable output (active low) enables front end encoder operation.
ERR-L	81	O	Error output (active low) from controller indicating a failure during message processing.
MS4-L	82	O	Machine state 4. Sequencer output used to enable logic internal to the hybrid. Test point for hybrid and module troubleshooting.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
LOAD-L	83	O	Sequencer output which during data transfers, indicates the module is reading from subsystem memory. Test point for hybrid and module troubleshooting.
SYNC	85	O	Sequencer output which during data transfers, indicates the word being transferred (read or write) is associated with a 1553 data word. Inverse of SNCS-L output. Test point for hybrid and module troubleshooting.
RPLY-L	86	O	Sequencer output which indicates the controller has transmitted the 1553 message and is in the reply mode waiting for the response from the remote terminal. Will stay low until RST1-L is asserted or one 6MHz-L clock pulse after CENA is driven low. Test point for hybrid and module troubleshooting.
R2/CE-L	72	O	Reset 2/FERST output (active low) activated by RST2-L or cycle enable low.
BCA-L	66	O	Bus controller active signal (active low) indicates the bus controller is processing a message.
BCTO-L	67	O	Bus controller timeout signal (active low) indicates the controller did not receive a response from the RT within the allowed response time. This is the signal affected by TB0-TB7.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
BINC-L	14	O	BUSINCR output (active low). Used to increment bus address counter at end of transfer cycle.
CWB0	30	O	Control bit 0. LSB of control word. Low indicates most recent command was an RT/RT transfer.
CWB1	29	O	Control bit 1. Low indicates most recent command was a broadcast command.
CWB2	28	O	Bit 2 of most recent control word. Low indicates most recent command is a mode code type.
CWB3	32	O	Control bit 3. Reserved.
CWB4	36	O	Control bit 4. Reserved.
CWB5	35	O	Control bit 5. Reserved.
CWB6	34	O	Control bit 6. May be used to initiate self-test of front end in conjunction with CWB1. Controller will initiate broadcast command and return to idle.
STS-L	39	O	Status set output (active low). Low indicates an address mismatch or status bit set in received status word.
LWO	37	O	Last word output. High indicates last word of message has been transmitted or received. Valid only during non-mode commands.
D0	43	I	Bit 0 of 16 bit data bus, LSB.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
D1	44	I	Bit 1 of 16 bit data bus.
D2	45	I	Bit 2 of 16 bit data bus.
D3	46	I	Bit 3 of 16 bit data bus.
D4	47	I	Bit 4 of 16 bit data bus.
D5	48	I	Bit 5 of 16 bit data bus.
D6	49	I	Bit 6 of 16 bit data bus.
D7	50	I	Bit 7 of 16 bit data bus.
D8	100	I	Bit 8 of 16 bit data bus.
D9	99	I	Bit 9 of 16 bit data bus.
D10	98	I	Bit 10 of 16 bit data bus.
D11	97	I	Bit 11 of 16 bit data bus.
D12	96	I	Bit 12 of 16 bit data bus.
D13	95	I	Bit 13 of 16 bit data bus.
D14	94	I	Bit 14 of 16 bit data bus.
D15	93	I	Bit 15 of 16 bit data bus, MSB.
NW-L	26	I	New word input (active low). Low indicates front end has new word in receiver buffer.
CMD-L	78	I	Command input (active low) from front end indicating sync type of received word. Low for command or status.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
VW-L	79	I	Valid word input (active low) from front end indicates word received was valid.
ND-L	22	I	No data input (active low) from front end. Low when 1553 bus is inactive.
TD-L	68	I	Transmitting data input (active low) from front end indicates encoder is shifting data to serial bus.
TACT-L	73	I	Transmitter active input (active low) from front end.
ANBR-L	27	I	Channel A new bus request input (active low). Low indicates first word received of new message on channel A.
BNBR-L	76	I	Channel B new bus request (active low) from front end.
CENA	21	I	Cycle enable input must be high to enable bus controller to process message transfer. When signal is low, hybrid is held in reset.
RST2-L	71	I	Reset 2 input (active low) from bus 67002. Activates R2/CE-L (active low). Connect to RST1-L when 67002 is not present.
CLKI	91	I	TTL clock input, divided by 2 to provide system clock for controller. Normally 12 MHz.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
TB0	65	I	Least significant timer bit. This is the 8 bit programmable timer input used for the no response timeout. Timeout period is selectable in 166 nS increments to allow for different RT response times.
TB1	64	I	Timer bit 1.
TB2	63	I	Timer bit 2.
TB3	62	I	Timer bit 3.
TB4	19	I	Timer bit 4.
TB5	18	I	Timer bit 5.
TB6	17	I	Timer bit 6.
TB7	16	I	Timer bit 7. MSB of timer input.
RST1-L	70	I	Reset 1 input (active low) should be connected to system reset. Resets bus controller.
BG-L	20	I	Bus grant input (active low). Handshake response from subsystem to BR output (active low). Gives bus controller access to parallel bus.
SEL1	69	I	Select input determines mode of operation for bus controller. When low, BC operates in 16 bit mode, when high 8 bit mode is used.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
BCEN-L	80	I	Bus controller enable input (active low). Low enables bus controller. High disables controller and tri-states the following bus control outputs: LSBS-L,MSBS-L,BDIR-L,BCA-L,LSBL-L, MSBL-L,LSBR-L,MSBR-L,LSBE-L, MSBE-L,LSBW-L,MSBW-L,BA1-BA6, T/R-L,CA-L/CB,BR-L,BGA-L,SNCS-L, TXEN-L,MS4-L,LOAD-L,SYNC,RPLY-L, ERR-L,ECEN-L.
CLKO	41	O	TTL compatible output from internal oscillator. Frequency controlled by crystal Y1, normally 12 MHz.
6MHz-L	42	O	TTL compatible output from 12 MHz oscillator divided by 2. Used during test.
Vcc	51	-	+5 V dc (CLOCK).
Vcc	1	-	+5 V dc.
Vcc	31	-	+5 V dc.
GND	5	-	Ground return.
GND	10	-	Ground return.
GND	15	-	Ground return.
GND	25	-	Ground return.
GND	40	-	Ground return.
GND	90	-	Ground return.
FRAME GND	11	-	Frame ground.

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6.4.3 Functional description of I/O pins - Continued.

Functional designation	Pin number	I/O	Functional description
		-	Unused pins: 52,77,87,88,89,92

6.5 Module handling consideration. Care should be exercised when using the UKT, UKU, and UKS modules to prevent bus contention during normal operation or power-up conditions. During software debugging where bus contentions are possible, current limiting of power supplies is a suggested method to prevent damage to modules.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N726)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Setup conditions: Connect A to pin	Setup conditions: Connect B to pin	25 ± 5°C Initial limits: Min	25 ± 5°C Initial limits: Max	Units	Notes
A1	Isolation, frame	C	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused pins	C,11	52,77, 87,88, 89,92				
A3	Isolation, unused to unused pins	52	77,87, 88,89, 92				
A4	Isolation, unused to unused pins	77	87,88, 89,92				
A5	Isolation, unused to unused pins	87	88,89, 92				
A6	Isolation, unused to unused pins	88	89,92				
A7	Isolation, unused to unused pins	89	92	10	INF	MOhm	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1
Megohm bridge voltage	30 V	± 10 percent	A2-A7
Isolation measurement	NA	± 500 KOhm	A1-A7
Isolation measurement (see note 6)	NA	± 50 KOhm	A1-A7

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins, except pin 11, shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Setup conditions: Connect A to pin	Setup conditions: Connect B to pin	25 ± 5°C Initial limits: Min	25 ± 5°C Initial limits: Max	Units	Notes
B1	Continuity, Vcc	1	31	0	1	Ohm	1-3,5
B2	Continuity, ground	5	10,15, 25,40,90				
B3	Continuity, ground	10	15,25, 40,90				
B4	Continuity, ground	15	25,40, 90				
B5	Continuity, ground	25	40,90				
B6	Continuity, ground	40	90	0	1	Ohm	1-3,5
B7	Continuity, frame	11	Frame	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

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NOTES: - Continued.

4. When performing test condition B7, penetrate the anodized finish on the module fin with a sharply pointed probe.

5. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Test description	Setup conditions Connect A to	Connect B to	25 ± 5°C Initial limits	Notes
C	Vcc stress test	1,31,51	5,10,15, 25,40,90	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.8 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. In the setup conditions in table IV, 'A' shall correspond to the positive terminal of the power supply, and 'B' shall correspond to the negative terminal.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	D0	43	5.5	0	100	uA	1-3
D2	D1	44					
D3	D2	45					
D4	D3	46					
D5	D4	47					
D6	D5	48					
D7	D6	49					
D8	D7	50					
D9	D8	100					
D10	D9	99					
D11	D10	98					
D12	D11	97					
D13	D12	96					
D14	D13	95					
D15	D14	94					
D16	D15	93					
D17	NW-L	26	5.5	0	100	uA	1-3
D18	CMD-L	78	5.5	0	1000	uA	1-3
D19	VW-L	79	5.5	0	1000	uA	1-3
D20	ND-L	22	5.5	0	100	uA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	TD-L	68	5.5	0	100	uA	1-3
D22	TACT-L	73					
D23	ANBR-L	27					
D24	BNBR-L	76					
D25	CENA	21					
D26	RST2-L	71					
D27	CLKI	91					
D28	TB0	65					
D29	TB1	64					
D30	TB2	63					
D31	TB3	62					
D32	TB4	19					
D33	TB5	18					
D34	TB6	17					
D35	TB7	16					
D36	RST1-L	70					
D37	BG-L	20					
D38	SEL1	69					
D39	BCEN-L	80	5.5	0	100	uA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	D1-D39
(Vcc,CLOCK)	5.5 V	± 55 mV	D1-D39
Power supply current limit,			
(Vcc	1.5 A	NA	D1-D39
(Vcc,CLOCK)	38.0 mA	NA	D1-D39
Forcing voltage,			
(Vin)	5.5 V	± 55 mV	D1-D39
Input current measurement	NA	± 5 uA	D1-D17
	NA	± 50 uA	D18,D19
	NA	± 5 uA	D20-D39

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	D0	43	-18.0	-1.5	-0.3	V	1-3
E2	D1	44					
E3	D2	45					
E4	D3	46					
E5	D4	47					
E6	D5	48					
E7	D6	49					
E8	D7	50					
E9	D8	100					
E10	D9	99					
E11	D10	98					
E12	D11	97					
E13	D12	96					
E14	D13	95					
E15	D14	94					
E16	D15	93					
E17	NW-L	26					
E18	CMD-L	78					
E19	VW-L	79					
E20	ND-L	22					
E21	TD-L	68	-18.0	-1.5	-0.3	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	TACT-L	73	-18.0	-1.5	-0.3	V	1-3
E23	ANBR-L	27					
E24	BNBR-L	76					
E25	CENA	21					
E26	RST2-L	71					
E27	CLKI	91					
E28	TB0	65					
E29	TB1	64					
E30	TB2	63					
E31	TB3	62					
E32	TB4	19					
E33	TB5	18					
E34	TB6	17					
E35	TB7	16					
E36	RST1-L	70					
E37	BG-L	20					
E38	SEL1	69					
E39	BCEN-L	80	-18.0	-1.5	-0.3	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	4.5 V	± 45 mV	E1-E39
(Vcc,CLOCK)	4.5 V	± 45 mV	E1-E39
Power supply current limit,			
(Vcc)	1.5 A	NA	E1-E39
(Vcc,CLOCK)	38.0 mA	NA	E1-E39
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E39
Input voltage measurement	NA	± 60 mV	E1-E39

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	LSBS-L	61	0.0	-130	-30	mA	1-4
F2	MSBS-L	9					
F3	BDIR-L	13					
F4	BCAC-L	12	0.0	-130	-30	mA	1-4
F5	CA-L/CB	33	0.0	-225	-40	mA	1-4
F6	SNCS-L	24					
F7	TXEN-L	23					
F8	MSBL-L	60					
F9	MSBE-L	58					
F10	LSBL-L	56					
F11	LSBE-L	7					
F12	T/R-L	38					
F13	LSBR-L	59					
F14	MSBR-L	8					
F15	LSBW-L	57					
F16	MSBW-L	6					
F17	BA1	53					
F18	BA2	2					
F19	BA3	3					
F20	BA4	55					
F21	BA5	4	0.0	-225	-40	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	BA6	54	0.0	-225	-40	mA	1-4
F23	BGA-L	74					
F24	BR-L	75	0.0	-225	-40	mA	1-4
F25	ECEN-L	84	0.0	-130	-30	mA	1-4
F26	ERR-L	81					
F27	MS4-L	82					
F28	LOAD-L	83					
F29	SYNC	85					
F30	RPLY-L	86	0.0	-130	-30	mA	1-4
F31	R2/CE-L	72	0.0	-100	-20	mA	1-4
F32	BCA-L	66					
F33	BCTO-L	67					
F34	BINC-L	14					
F35	CWB0	30					
F36	CWB1	29					
F37	CWB2	28					
F38	CWB3	32					
F39	CWB4	36					
F40	CWB5 1/	35	0.0	-100	-20	mA	1-4

1/ Output pin CWB5 is connected to 2 outputs internal to the module.

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	CWB6	34	0.0	-100	-20	mA	1-4
F42	STS-L	39	|	|	|	|	|
F43	LWO	37	0.0	-100	-20	mA	1-4
F44	6MHz-L	42	0.0	-225	-40	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	F1-F44
(Vcc,CLOCK)	5.5 V	± 55 mV	F1-F44
Power supply current limit,			
(Vcc)	1.5 A	NA	F1-F44
(Vcc,CLOCK)	38.0 mA	NA	F1-F44
Forcing voltage,			
(Vout)	0.0 V	± 50 mV	F1-F44
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F44
low, (VL)	0.0 V	± 100 mV	F1-F44
Output current measurement	NA	± 5.0 mA	F1-F4
	NA	± 9.25 mA	F5-F24
	NA	± 5.0 mA	F25-F30
	NA	± 4.0 mA	F31-F39
	NA	± 9.25 mA	F40
	NA	± 4.0 mA	F41-F43
	NA	± 5.0 mA	F44

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	D0	43	2.7	-2	20	uA	1-3
G2	D1	44					
G3	D2	45					
G4	D3	46					
G5	D4	47					
G6	D5	48					
G7	D6	49					
G8	D7	50					
G9	D8	100					
G10	D9	99					
G11	D10	98					
G12	D11	97					
G13	D12	96					
G14	D13	95					
G15	D14	94					
G16	D15	93					
G17	NW-L	26	2.7	-2	20	uA	1-3
G18	CMD-L	78	2.7	-2	25	uA	1-3
G19	VW-L	79	2.7	-2	50	uA	1-3
G20	ND-L	22	2.7	-2	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G21	TD-L	68	2.7	-2	20	uA	1-3
G22	TACT-L	73	2.7	-2	20	uA	1-3
G23	ANBR-L	27	2.7	-200	-50	uA	1-3
G24	BNBR-L	76	2.7	-200	-50	uA	1-3
G25	CENA	21	2.7	-2	20	uA	1-3
G26	RST2-L	71	2.7	-200	-50	uA	1-3
G27	CLKI	91	2.7	-2	20	uA	1-3
G28	TB0	65					
G29	TB1	64					
G30	TB2	63					
G31	TB3	62					
G32	TB4	19					
G33	TB5	18					
G34	TB6	17					
G35	TB7	16					
G36	RST1-L	70					
G37	BG-L	20	2.7	-2	20	uA	1-3
G38	SEL1	69	2.7	-2	100	uA	1-3
G39	BCEN-L	80	2.7	-2	150	uA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	G1-G39
(Vcc,CLOCK)	5.5 V	± 55 mV	G1-G39
Power supply current limit,			
(Vcc)	1.5 A	NA	G1-G39
(Vcc,CLOCK)	38.0 mA	NA	G1-G39
Forcing voltage,			
(Vih)	2.7 V	± 50 mV	G1-G39
Input current measurement	NA	± 1.10 uA	G1-G17
	NA	± 1.35 uA	G18
	NA	± 2.60 uA	G19
	NA	± 1.10 uA	G20-G22
	NA	± 7.50 uA	G23,G24
	NA	± 1.10 uA	G25
	NA	± 7.50 uA	G26
	NA	± 1.10 uA	G27-G37
	NA	± 5.10 uA	G38
	NA	± 7.60 uA	G39

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	D0	43	0.4	-0.2	0.02	mA	1-3
H2	D1	44					
H3	D2	45					
H4	D3	46					
H5	D4	47					
H6	D5	48					
H7	D6	49					
H8	D7	50	0.4	-0.2	0.02	mA	1-3
H9	D8	100	0.4	-0.4	0.02	mA	1-3
H10	D9	99					
H11	D10	98					
H12	D11	97					
H13	D12	96					
H14	D13	95					
H15	D14	94					
H16	D15	93					
H17	NW-L	26					
H18	CMD-L	78					
H19	VW-I	79					
H20	ND-L	22					
H21	TD-L	68	0.4	-0.4	0.02	mA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	TACT-L	73	0.4	-0.4	0.02	mA	1-3
H23	ANBR-L	27	0.4	-0.8	-0.2	mA	1-3
H24	BNBR-L	76	0.4	-0.8	-0.2	mA	1-3
H25	CENA	21	0.4	-0.4	0.02	mA	1-3
H26	RST2-L	71	0.4	-0.8	-0.2	mA	1-3
H27	CLKI	91	0.4	-0.4	0.02	mA	1-3
H28	TB0	65					
H29	TB1	64					
H30	TB2	63					
H31	TB3	62					
H32	TB4	19					
H33	TB5	18					
H34	TB6	17					
H35	TB7	16					
H36	RST1-L	70					
H37	BG-L	20	0.4	-0.4	0.02	mA	1-3
H38	SEL1	69	0.4	-1.6	0.02	mA	1-3
H39	BCEN-L	80	0.4	-1.6	0.02	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	H1-H39
(Vcc,CLOCK)	5.5 V	± 55 mV	H1-H39
Power supply current limit,			
(Vcc)	1.5 A	NA	H1-H39
(Vcc,CLOCK)	38.0 mA	NA	H1-H39
Forcing voltage,			
(Vil)	0.4 V	± 50 mV	H1-H39
Input current measurement	NA	± 11 uA	H1-H8
	NA	± 21 uA	H9-H22
	NA	± 30 uA	H23,H24
	NA	± 21 uA	H25
	NA	± 30 uA	H26
	NA	± 21 uA	H27-H37
	NA	± 81 uA	H38,H39

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(c) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	LSBS-L	61	-1.0	2.4	4.5	V	1-5
J2	MSBS-L	9					
J3	BDIR-L	13					
J4	BCAC-L	12	-1.0	2.4	4.5	V	1-5
J5	CA-L/CB	33	-3.0	2.4	4.5	V	1-5
J6	SNCS-L	24					
J7	TXEN-L	23					
J8	MSBL-L	60					
J9	MSBE-L	58					
J10	LSBL-L	56					
J11	LSBE-L	7					
J12	T/R-L	38					
J13	LSBR-L	59					
J14	MSBR-L	8					
J15	LSBW-L	57					
J16	MSBW-L	6					
J17	BA1	53					
J18	BA2	2					
J19	BA3	3					
J20	BA4	55					
J21	BA5	4	-3.0	2.4	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	BA6	54	-3.0	2.4	4.5	V	1-5
J23	BGA-L	74					
J24	BR-L	75	-3.0	2.4	4.5	V	1-5
J25	ECEN-L	84	-0.8	2.4	4.5	V	1-5
J26	ERR-L	81					
J27	MS4-L	82					
J28	LOAD-L	83					
J29	SYNC	85					
J30	RPLY-L	86	-0.8	2.4	4.5	V	1-5
J31	R2/CE-L	72	-0.4	2.4	4.5	V	1-5
J32	BCA-L	66					
J33	BCTO-L	67					
J34	BINC-L	14					
J35	CWB0	30					
J36	CWB1	29					
J37	CWB2	28					
J38	CWB3	32					
J39	CWB4	36					
J40	CWB5	35					
J41	CWB6	34	-0.4	2.4	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J42	STS-L	39	-0.4	2.4	4.5	V	1-5
J43	LWO	37	-0.4	2.4	4.5	V	1-5
J44	CLKO	41	-10.0	4.1	4.4	V	1-5
J45	6MHz-L	42	-3.0	2.4	4.5	V	1-5

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	4.5 V	± 45 mV	J1-J45
(Vcc,CLOCK)	4.5 V	± 45 mV	J1-J45
Power supply current limit,			
(Vcc)	1.5 A	NA	J1-J45
(Vcc,CLOCK)	38.0 mA	NA	J1-J45
Forcing current,			
(Ioh)	-1.0 mA	± 10 uA	J1-J4
	-3.0 mA	± 30 uA	J5-J24
	-0.8 mA	± 8 uA	J25-J30
	-0.4 mA	± 4 uA	J31-J43
	-10.0 mA	± 100 uA	J44
	-3.0 mA	± 30 uA	J45
Pattern voltage			
high, (VH)	2.0 V	± 50 mV	J1-J45
low, (VL)	0.7 V	± 50 mV	J1-J45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage measurement	NA NA NA	± 105 mV ± 20 mV ± 105 mV	J1-J43 J44 J45

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

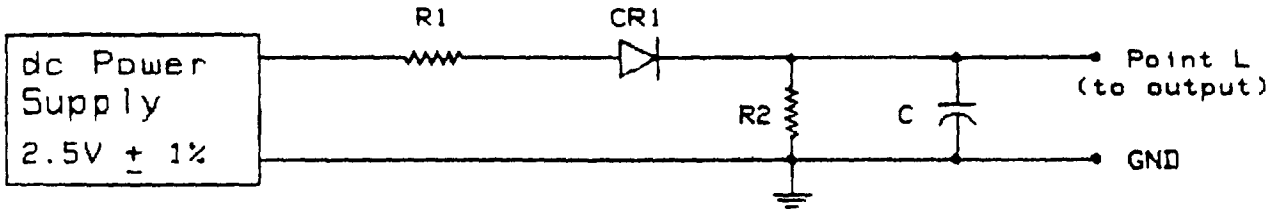
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input shall be verified.

4. The load shall be as indicated on figure 18.

5. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4150 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for test conditions P and Q.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions T and U.

(c) C not required for dc tests.

FIGURE 18. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
61,9,13,12	2430	12.0	0.4
33,24,23,60,58, 56,7,38,59,8,57, 53,2,3,55,4,54, 74,75,6	806	12.0	0.4
42	806	8.0	0.5
72,66,67,14,30, 29,32,36,34,39, 37,28	6340	4.0	0.4
35	6040	4.0	0.4
41	412	10.0	0.5
84,81,82,83,85,86	1210	8.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/8 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	LSBS-L	61	12.0	0.0	0.4	V	1-5
K2	MSBS-L	9					
K3	BDIR-L	13					
K4	BCAC-L	12					
K5	CA-L/CB	33					
K6	SNCS-L	24					
K7	TXEN-L	23					
K8	MSBL-L	60					
K9	MSBE-L	58					
K10	LSBL-L	56					
K11	LSBE-L	7					
K12	T/R-L	38					
K13	LSBR-L	59					
K14	MSBR-L	8					
K15	LSBW-L	57					
K16	MSBW-L	6					
K17	BA1	53					
K18	BA2	2					
K19	BA3	3					
K20	BA4	55	12.0	0.0	0.4	V	1-5

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Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	BA5	4	12.0	0.0	0.4	V	1-5
K22	BA6	54					
K23	BGA-L	74					
K24	BR-L	75	12.0	0.0	0.4	V	1-5
K25	ECEN-L	84	8.0	0.0	0.5	V	1-5
K26	ERR-L	81					
K27	MS4-L	82					
K28	LOAD-L	83					
K29	SYNC	85					
K30	RPLY-L	86	8.0	0.0	0.5	V	1-5
K31	R2/CE-L	72	4.0	0.0	0.4	V	1-5
K32	BCA-L	66					
K33	BCTO-L	67					
K34	BINC-L	14					
K35	CWB0	30					
K36	CWB1	29					
K37	CWB2	28					
K38	CWB3	32					
K39	CWB4	36					
K40	CWB5	35					
K41	CWB6	34	4.0	0.0	0.4	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K42	STS-L	39	4.0	0.0	0.4	V	1-5
K43	LWO	37	4.0	0.0	0.4	V	1-5
K44	CLKO	41	10.0	0.0	0.5	V	1-5
K45	6MHz-L	42	8.0	0.0	0.4	V	1-5

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	4.5 V	± 45 mV	K1-K45
(Vcc,CLOCK)	4.5 V	± 45 mV	K1-K45
Power supply current limit,			
(Vcc)	1.5 A	NA	K1-K45
(Vcc,CLOCK)	38.0 mA	NA	K1-K45
Forcing current,			
(Iol)	12.0 mA	± 120 uA	K1-K24
	8.0 mA	± 80 uA	K25-K30
	4.0 mA	± 40 uA	K31-K43
	10.0 mA	± 100 uA	K44
	8.0 mA	± 80 uA	K45
Pattern voltage,			
high, (VH)	2.0 V	± 50 mV	K1-K45
low, (VL)	0.7 V	± 50 mV	K1-K45

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NOTES: - Continued.

Condition	Value	Accuacy	Test condition
Output voltage measurement	NA	± 20 mV	K1-K24
	NA	± 25 mV	K25-K30
	NA	± 20 mV	K31-K43
	NA	± 25 mV	K44
	NA	± 20 mV	K45

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. During this test, the high and low level input threshold voltage of every input shall be verified. This may require redundant output voltage tests on some pins.

4. The load shall be as indicated on figure 18.

5. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	LSBS-L	61	2.4	-20	20	uA	1-3
L2	MSBS-L	9					
L3	BDIR-L	13					
L4	BCAC-L	12					
L5	CA-L/CB	33					
L6	SNCS-L	24					
L7	TXEN-L	23	2.4	-20	20	uA	1-3
L8	MSBL-L	60	2.4	-225	-100	uA	1-3
L9	MSBE-L	58					
L10	LSBL-L	56					
L11	LSBE-L	7	2.4	-225	-100	uA	1-3
L12	T/R-L	38	2.4	-20	20	uA	1-3
L13	LSBR-L	59	2.4	-225	-100	uA	1-3
L14	MSBR-L	8					
L15	LSBW-L	57					
L16	MSBW-L	6	2.4	-225	-100	uA	1-3
L17	BA1	53	2.4	-20	20	uA	1-3
L18	BA2	2					
L19	BA3	3					
L20	BA4	55					
L21	BA5	4	2.4	-20	20	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Voh (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L22	BA6	54	2.4	-20	20	uA	1-3
L23	BGA-L	74					
L24	BR-L	75	2.4	-20	20	uA	1-3
L25	ECEN-L	84	2.4	-750	145	uA	1-3
L26	ERR-L	81	2.4	-100	100	uA	1-3
L27	MS4-L	82	2.4	-350	125	uA	1-3
L28	LOAD-L	83	2.4	-1950	205	uA	1-3
L29	SYNC	85	2.4	-1950	225	uA	1-3
L30	RPLY-L	86	2.4	-2950	285	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	L1-L30
(Vcc,CLOCK)	5.5 V	± 55 mV	L1-L30
Power supply current limit,			
(Vcc)	1.5 A	NA	L1-L30
(Vcc,CLOCK)	38.0 mA	NA	L1-L30

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:08Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Voh)	2.4 V	± 50 mV	L1-L30
Pattern voltage, high, (VH)	3.0 V	± 100 mV	L1-L30
low, (VL)	0.0 V	± 100 mV	L1-L30
Output current measurement	NA	± 2.00 uA	L1-L7
	NA	± 6.25 uA	L8-L11
	NA	± 2.00 uA	L12
	NA	± 6.25 uA	L13-L16
	NA	± 2.00 uA	L17-L24
	NA	± 44.75 uA	L25
	NA	± 10.00 uA	L26
	NA	± 23.75 uA	L27
	NA	± 107.75 uA	L28
	NA	± 108.75 uA	L29
	NA	± 161.75 uA	L30

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	LSBS-L	61	0.4	-20	20	uA	1-3
M2	MSBS-L	9					
M3	BDIR-L	13					
M4	BCAC-L	12					
M5	CA-L/CB	33					
M6	SNCS-L	24					
M7	TXEN-L	23	0.4	-20	20	uA	1-3
M8	MSBL-L	60	0.4	-350	-200	uA	1-3
M9	MSBE-L	58					
M10	LSBL-L	56					
M11	LSBE-L	7	0.4	-350	-200	uA	1-3
M12	T/R-L	38	0.4	-20	20	uA	1-3
M13	LSBR-L	59	0.4	-350	-200	uA	1-3
M14	MSBR-L	8					
M15	LSBW-L	57					
M16	MSBW-L	6	0.4	-350	-200	uA	1-3
M17	BA1	53	0.4	-20	20	uA	1-3
M18	BA2	2					
M19	BA3	3					
M20	BA4	55	0.4	-20	20	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M21	BA5	4	0.4	-20	20	uA	1-3
M22	BA6	54					
M23	BGA-L	74					
M24	BR-L	75	0.4	-20	20	uA	1-3
M25	ECEN-L	84	0.5	-750	145	uA	1-3
M26	ERR-L	81	0.5	-100	100	uA	1-3
M27	MS4-L	82	0.5	-350	125	uA	1-3
M28	LOAD-L	83	0.5	-1950	205	uA	1-3
M29	SYNC	85	0.5	-1950	225	uA	1-3
M30	RPLY-L	86	0.5	-2950	285	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	M1-M30
(Vcc,CLOCK)	5.5 V	± 55 mV	M1-M30
Power supply current limit,			
(Vcc)	1.5 A	NA	M1-M30
(Vcc,CLOCK)	38.0 mA	NA	M1-M30

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:08Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vol)	0.4 V	± 50 mV	M1-M24
	0.5 V	± 50 mV	M25-M30
Pattern voltage, high, (VH)	3.0 V	± 100 mV	M1-M30
low, (VL)	0.0 V	± 100 mV	M1-M30
Output current measurement	NA	± 2.00 uA	M1-M7
	NA	± 7.50 uA	M8-M11
	NA	± 2.00 uA	M12
	NA	± 7.50 uA	M13-M16
	NA	± 2.00 uA	M17-M24
	NA	± 44.75 uA	M25
	NA	± 10.00 uA	M26
	NA	± 23.75 uA	M27
	NA	± 107.75 uA	M28
	NA	± 108.75 uA	M29
	NA	± 161.75 uA	M30

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Icc	1,31	250	1000	mA	1-3
N2	Icc, CLOCK	51	2	25	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
(Vcc,CLOCK)	5.5 V	± 55 mV	N1,N2
Power supply current limit, (Vcc)	1.5 A	NA	N1,N2
(Vcc,CLOCK)	38.0 mA	NA	N1,N2
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1,N2
low, (VL)	0.0 V	± 100 mV	N1,N2
Power supply current measurement	NA	± 55.50 mA	N1
	NA	± 1.15 mA	N2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the modul
measurement at the pattern line
current as indicated in table X

3. Abbreviations used are defi

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TABLE XV. Functional test.

Test cond	Test description	Notes	Fig
P1	Functional test minimum power supply voltage maximum clock frequency	1-3	5-18
P2	Functional test maximum power supply voltage maximum clock frequency	1-3	5-18

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (Vcc,CLOCK) (Vcc) (Vcc,CLOCK)	 4.5 V 4.5 V 5.5 V 5.5 V	 ± 45 mV ± 45 mV ± 55 mV ± 55 mV	 P1 P1 P2 P2
Power supply current limit, (Vcc) (Vcc,CLOCK)	 1.5 A 38.0 mA	 NA NA	 P1,P2 P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 P1,P2 P1,P2

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	P1
high, (Voh)	NA	± 155 mV	P2
low, (Vol)	NA	± 20 mV	P1,P2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 18). The output compare levels shall be as follows:

High level: 2.4 V ≤ Voh ≤ Vcc; all pins except pin 41
Vcc-0.4 V ≤ Voh ≤ Vcc; pin 41
Low level: 0.0 V ≤ Vol ≤ 0.5 V; all pins

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes	Figs
Q1	Dynamic test, sample maximum clock frequency minimum power supply	1-6	5-18
Q2	Dynamic test, sample maximum clock frequency maximum power supply	1-6	5-18

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (Vcc,CLOCK) (Vcc) (Vcc,CLOCK)	 4.5 V 4.5 V 5.5 V 5.5 V	 ± 45 mV ± 45 mV ± 45 mV ± 55 mV	 Q1 Q1 Q2 Q2
Power supply current limit, (Vcc) (Vcc,CLOCK)	 1.5 A 38.0 mA	 NA NA	 Q1,Q2 Q1,Q2
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 Q1,Q2 Q1,Q2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:08Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1
high, (Voh)	NA	± 155 mV	Q2
low, (Vol)	NA	± 20 mV	Q1,Q2

2. The following procedure is required to perform this test:

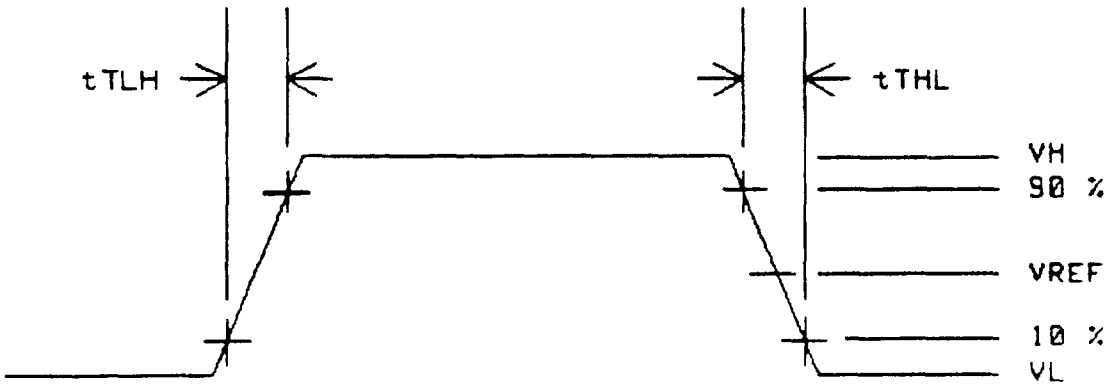
(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: 2.4 V ≤ Voh ≤ Vcc; all pins except pin 41
Vcc-0.4 V ≤ Voh ≤ Vcc; pin 41
Low level: 0.0 V ≤ Vol ≤ 0.5 V; all pins

(c) The dynamic parameters are found in 6.3.3.

3. Input/Output (I/O) waveforms are as specified on figures 5-17. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

4. The load circuit for this test is specified on figure 18 with CL = 150 pF.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	PSFC, Vcc	1,31	4.9	9.1	uF	1-3
R2	PSFC, Vcc (CLOCK)	51	0.007	0.013	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R1,R2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R1,R2
Capacitance measurement	NA NA	± 210 nF ± 300 pF	R1 R2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. Abbreviations used are defined in 6.1.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
S1	PSTA, Vcc	1,31	0	400	mV	1-3	19
S2	PSTA, Vcc (CLOCK)	51	0	400	mV	1-3	19

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S1,S2
(Vcc,CLOCK)	5.0 V	± 250 mV	S1,S2
Power supply current limit,			
(Vcc)	1.5 A	NA	S1,S2
(Vcc,CLOCK)	38.0 mA	NA	S1,S2
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	S1,S2
low, (VL)	0.0 V	± 100 mV	S1,S2
Power supply transient amplitude measurement	NA	± 20 mV	S1,S2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) All Vcc pins shall be shorted together and the scope point, Y01, shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D07, set at a maximum strobe rate of one-tenth of the cycle time per division, visually inspect each pattern row for the most negative spike as indicated on figure 19.

(e) The maximum probe capacitance is 10 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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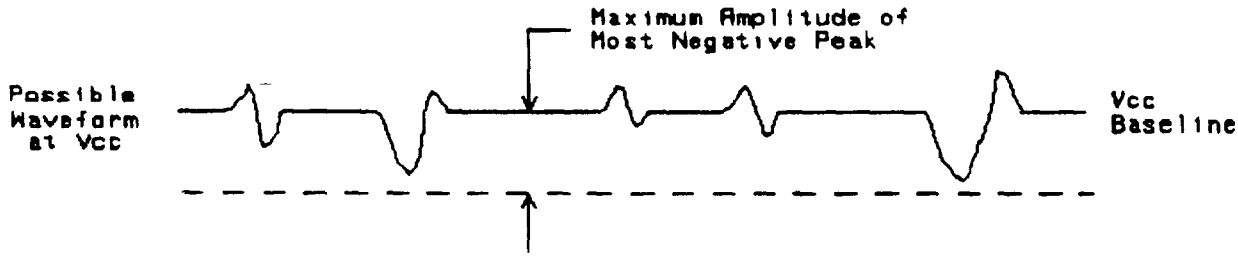


FIGURE 19. Power supply transient amplitude waveform for table XVIII.

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TABLE XIX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T1	LSBS-L	61	0	400	mV	1-5	18, 20, 21
T2	MSBS-L	9					
T3	BDIR-L	13					
T4	BCAC-L	12					
T5	CA-L/CB	33					
T6	SNCS-L	24					
T7	TXEN-L	23					
T8	MSBL-L	60					
T9	MSBE-L	58					
T10	LSBL-L	56					
T11	LSBE-L	7					
T12	T/R-L	38					
T13	LSBR-L	59					
T14	MSBR-L	8					
T15	LSBW-L	57					
T16	MSBW-L	6					
T17	BA1	53					
T18	BA2	2					
T19	BA3	3					
T20	BA4	55	0	400	mV	1-5	18, 20, 21

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TABLE XIX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T21	BA5	4	0	400	mV	1-5	18, 20, 21
T22	BA6	54					
T23	BGA-L	74					
T24	BR-L	75					
T25	ECEN-L	84					
T26	ERR-L	81					
T27	MS4-L	82					
T28	LOAD-L	83					
T29	SYNC	85					
T30	RPLY-L	86					
T31	R2/CE-L	72					
T32	BCA-L	66					
T33	BCTO-L	67					
T34	BINC-L	14					
T35	CWB0	30					
T36	CWB1	29					
T37	CWB2	28					
T38	CWB3	32					
T39	CWB4	36					
T40	CWB5	35	0	400	mV	1-5	18, 20, 21

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TABLE XIX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T41	CWB6	34	0	400	mV	1-5	18, 20, 21
T42	STS-L	39					
T43	LWO	37	0	400	mV	1-5	18, 20, 21

See notes following table XX.

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TABLE XX. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U1	LSBS-L	61	0	400	mV	1-5	18, 20, 21
U2	MSBS-L	9					
U3	BDIR-L	13					
U4	BCAC-L	12					
U5	CA-L/CB	33					
U6	SNCS-L	24					
U7	TXEN-L	23					
U8	MSBL-L	60					
U9	MSBE-L	58					
U10	LSBL-L	56					
U11	LSBE-L	7					
U12	T/R-L	38					
U13	LSBR-L	59					
U14	MSBR-L	8					
U15	LSBW-L	57					
U16	MSBW-L	6					
U17	BA1	53					
U18	BA2	2					
U19	BA3	3					
U20	BA4	55	0	400	mV	1-5	18, 20, 21

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TABLE XX. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U21	BA5	4	0	400	mV	1-5	18, 20, 21
U22	BA6	54					
U23	BGA-L	74					
U24	BR-L	75					
U25	ECEN-L	84					
U26	ERR-L	81					
U27	MS4-L	82					
U28	LOAD-L	83					
U29	SYNC	85					
U30	RPLY-L	86					
U31	R2/CE-L	72					
U32	BCA-L	66					
U33	BCTO-L	67					
U34	BINC-L	14					
U35	CWB0	30					
U36	CWB1	29					
U37	CWB2	28					
U38	CWB3	32					
U39	CWB4	36					
U40	CWB5	35	0	400	mV	1-5	18, 20, 21

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TABLE XX. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U41	CWB6	34	0	400	mV	1-5	18, 20, 21
U42	STS-L	39					
U43	LWO	37	0	400	mV	1-5	18, 20, 21

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NOTES:

1. Output crosstalk, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1-T43, U1-U43
(Vcc,CLOCK)	5.0 V	± 250 mV	T1-T43, U1-U43
Power supply current limit, (Vcc)	1.5 A	NA	T1-T43, U1-U43
(Vcc,CLOCK)	38.0 mA	NA	T1-T43, U1-U43
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1-T43, U1-U43
low, (VL)	0.0 V	± 100 mV	T1-T43, U1-U43
Output crosstalk measurement	NA	± 20 mV	T1-T43, U1-U43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in tables XIX and XX.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in tables XIX and XX, connect probe 'B' and visually inspect the output for the most positive or negative spike as indicated on figure 20.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 18 with CL = 50 pF.

4. This test circuit shall be as indicated on figure 21.

5. Abbreviations used are defined in 6.1.

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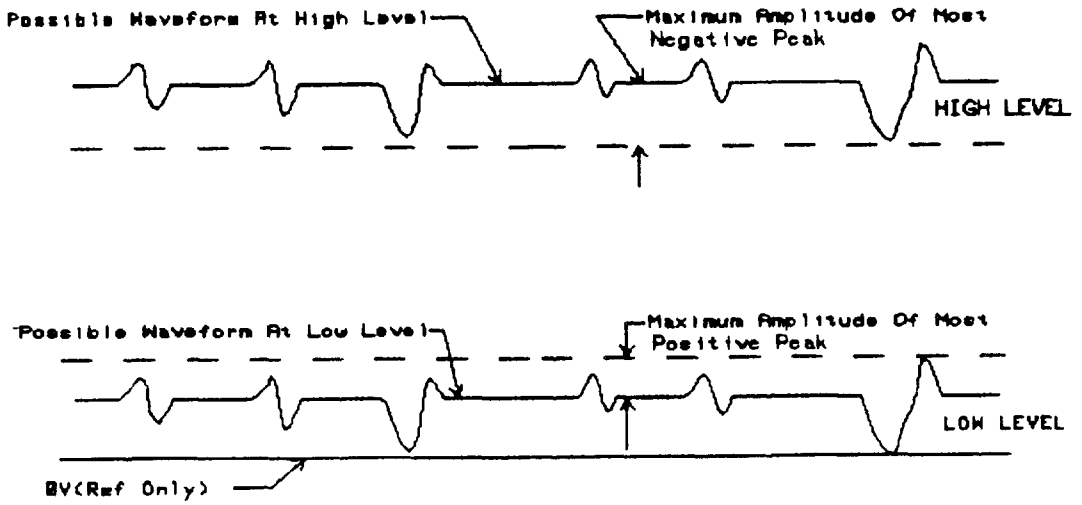


FIGURE 20. Output crosstalk test circuit waveforms, high and low level, for tables XIX and XX.

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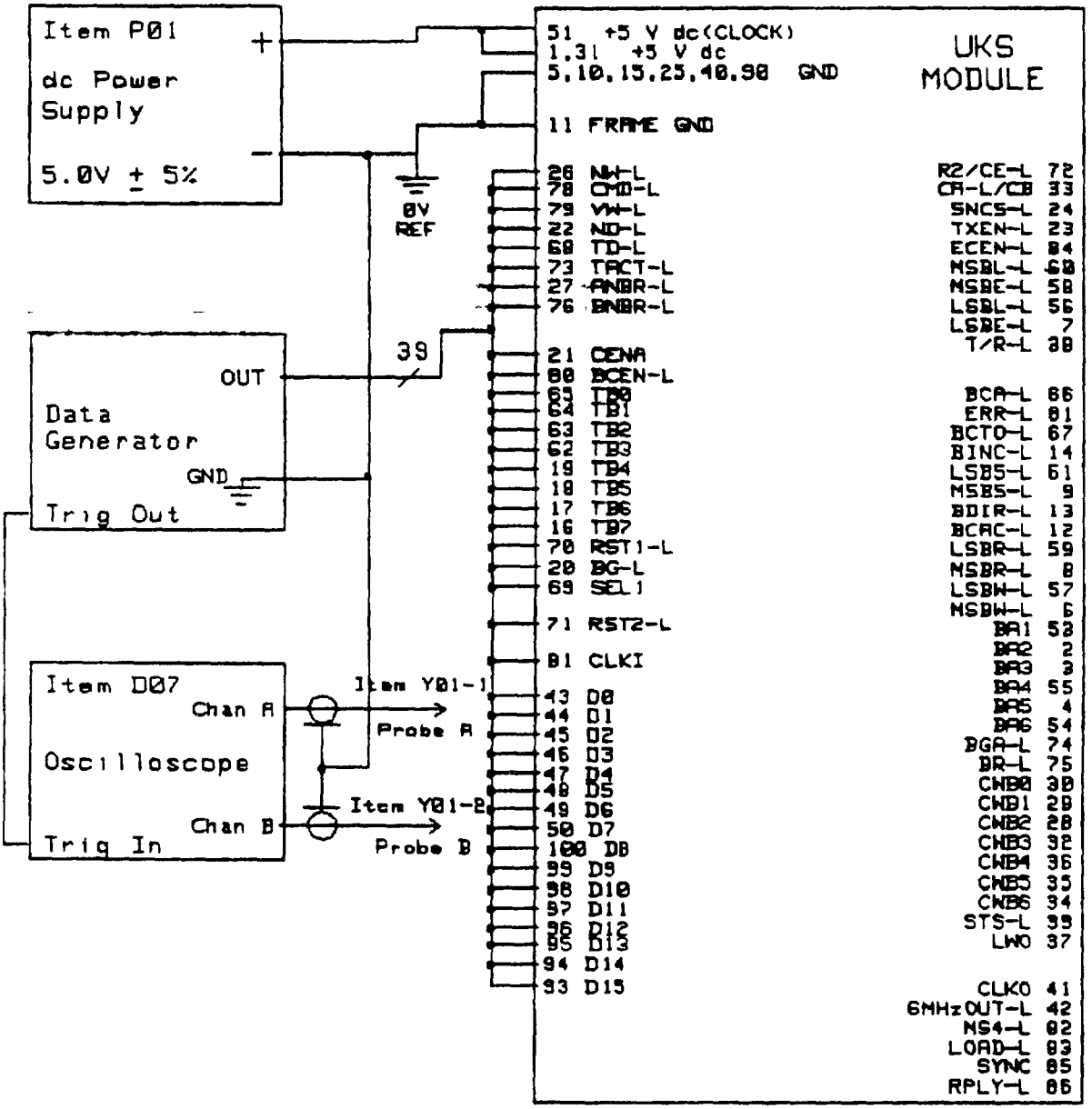


FIGURE 21. Output crosstalk, high and low level, test circuit for tables XIX and XX.

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TABLE XXI. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V1	D0	43	2	40	pF	1-3
V2	D1	44				
V3	D2	45				
V4	D3	46				
V5	D4	47				
V6	D5	48				
V7	D6	49				
V8	D7	50				
V9	D8	100				
V10	D9	99				
V11	D10	98				
V12	D11	97				
V13	D12	96				
V14	D13	95				
V15	D14	94				
V16	D15	93				
V17	NW-L	26				
V18	CMD-L	78				
V19	VW-L	79				
V20	ND-L	22				
V21	TD-L	68	2	40	pF	1-3

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TABLE XXI. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V22	TACT-L	73	2	40	pF	1-3
V23	ANBR-L	27				
V24	BNBR-L	76				
V25	CENA	21				
V26	RST2-L	71				
V27	CLKI	91				
V28	TB0	65				
V29	TB1	64				
V30	TB2	63				
V31	TB3	62				
V32	TB4	19				
V33	TB5	18				
V34	TB6	17				
V35	TB7	16				
V36	RST1-L	70				
V37	BG-L	20				
V38	SEL1	69	2	40	pF	1-3
V39	BCEN-L	80	2	75	pF	1-3

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	V1-V39
Capacitance meter voltage	15 mV	± 1.5 mV	V1-V39
Capacitance measurement	NA	± 1.90 pF ± 3.65 pF	V1-V39 V39

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXI.

3. Abbreviations used are defined in 6.1.

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TABLE XXII. Oscillator test (CLKO).

Test cond	Parameter	Connect A to pin	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
W1	PWH	41	28.0	55.33	nS	1-5	22-24
W2	PWL	41	28.0	55.33	nS	1-5	22-24
W3	PRR	41	83.25	83.42	nS	1-5	22-24
W4	Voh	41	5.0	--	V	1-5	22-24
W5	Vol	41	--	0.4	V	1-5	22-24

NOTES:

1. Oscillator test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply,			
(Vcc)	5.5 V	± 55 mV	W1-W5
(Vcc, CLOCK)	5.5 V	± 55 mV	W1-W5
Power supply current limit,			
(Vcc)	1.5 A	NA	W1-W5
(Vcc, CLOCK)	38.0 mA	NA	W1-W5
Oscillator test measurements	NA	± 1.4 uS	W1,W2
	NA	± 0.09 nS	W3
	NA	± 250.0 mV	W4
	NA	± 20.0 mV	W5

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as indicated on figure 22. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

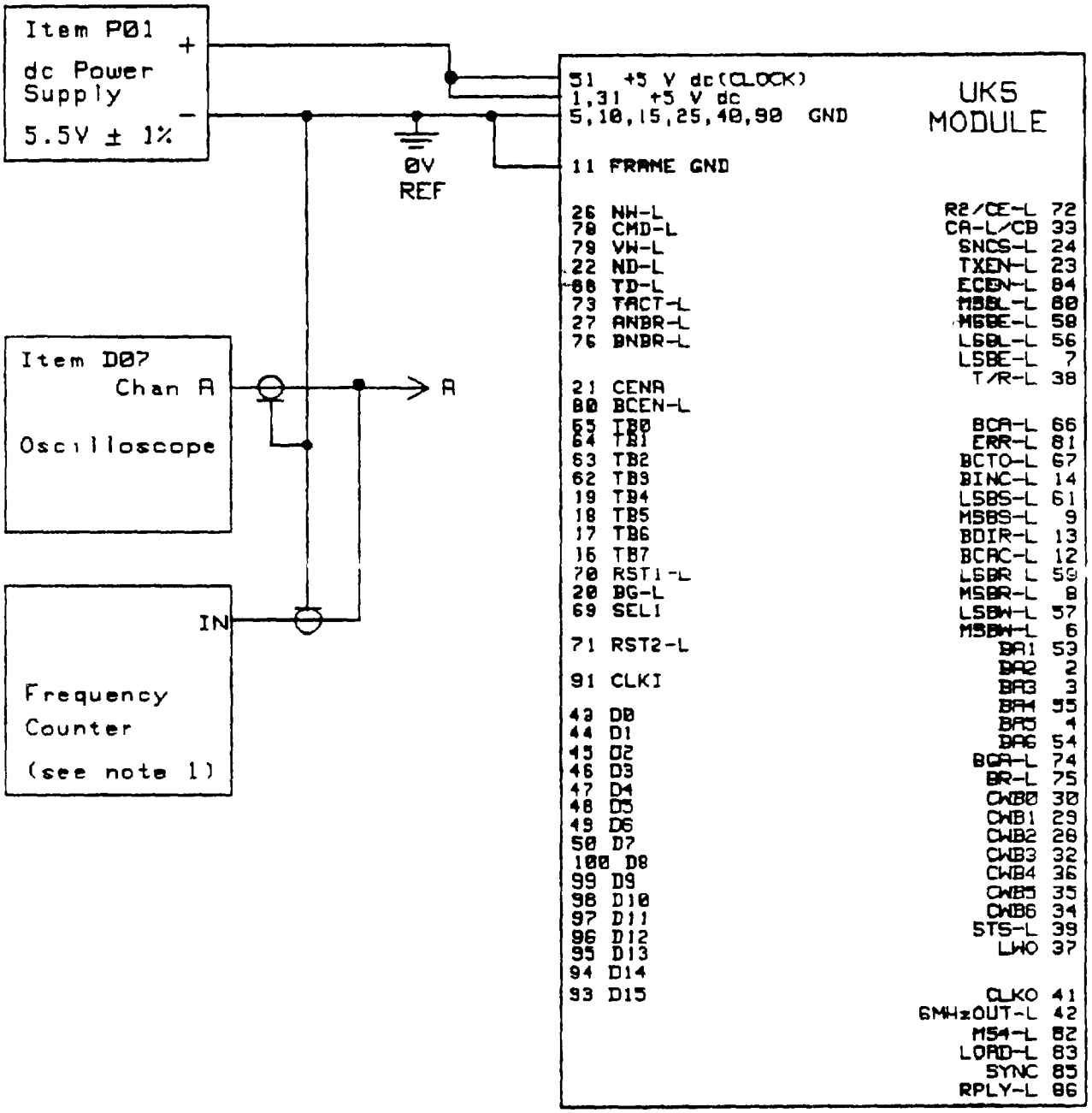
(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Pin 41 shall be loaded as indicated on figure 23.

4. Take the measurements indicated on figure 24.

5. Abbreviations used are defined in 6.1.

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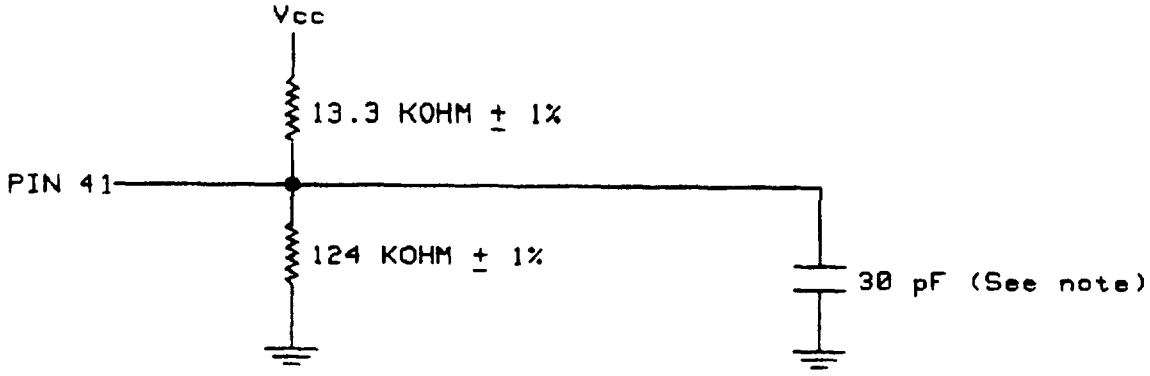


NOTE:

1. Use HP5315A or equivalent for test condition W3.

FIGURE 22. Oscillator test circuit.

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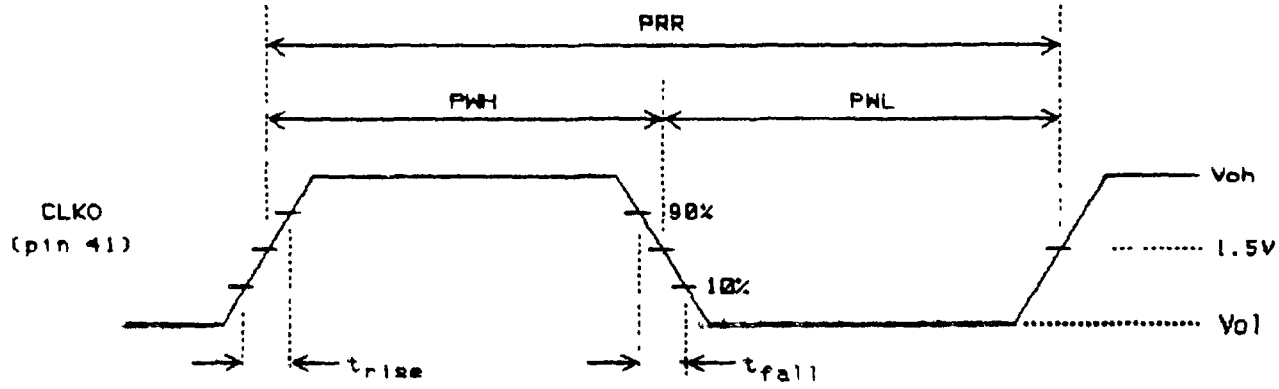


NOTE:

1. Includes probe capacitance.

FIGURE 23. Load circuit for pin 41.

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NOTE:

1. When measuring output, ignore any overshoot or undershoot. Take readings from the center of pulses.

FIGURE 24. Oscillator test circuit waveform.

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(See Instructions – Reverse Side)

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b. Recommended Wording

c. Reason/Rationale for Recommendation

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